

Online Appendix for “Understanding Disparities in Punishment: Regulator Preferences and Expertise”

Karam Kang and Bernardo S Silveira

APPENDIX A. REVISITING MODELING ASSUMPTIONS

A.1. The 2006 Institutional Changes and Confounding Factors. Our research design relies on the changes in compliance and enforcement before and after the 2006 institutional changes, conditional on (time-varying) observed attributes. Section 3.3 documents unconditional time trends in compliance and enforcement (Figure 1), and the present section shows that these trends persist even after controlling for all observed facility and local attributes.

Column (1) of Table A1 provides the regression results of the following equation:

$$compliance_{i,q,y} = \beta \mathbf{x}_{i,q,y} + \mu_i + \delta_q + \phi_y + \xi_{i,q,y},$$

where $compliance_{i,q,y}$ is a dummy indicating that no effluent MMP violations occurred in facility i during the q^{th} quarter of year y , and $\mathbf{x}_{i,q,y}$ is a vector of all facility and local attributes delineated in Table 1. Panel (A) in Figure A1 shows the regression coefficient estimates of year dummies (ϕ_y) with the year of 2000 being the base year. The estimated year dummy coefficients are not statistically different from zero for the years of 2001–2007, while those after 2007 are significant and are greater than zero. If we interpret these year dummy coefficients after 2006 as the effects of the 2006 institutional changes on compliance, then we conclude that there is a two-year lag.

Columns (2)–(4) of Table A1 provides the regression results of the following equation, and Panel (B) in Figure A1 shows the regression coefficient estimates of year dummies ($\tilde{\phi}_y$) presented in Column (2):

$$\log(penalty_{v,i,q,y} + 1) = \alpha \mathbf{z}_{v,i,q,y} + \tilde{\beta} \mathbf{x}_{i,q,y} + \tilde{\mu}_i + \tilde{\delta}_q + \tilde{\phi}_y + \tilde{\epsilon}_{v,i,q,y},$$

where $penalty_{v,i,q,y}$ is the amount of penalty imposed within four years of the occurrence date of violation v by facility i during the q^{th} quarter of year y , and $\mathbf{z}_{v,i,q,y}$ denotes the violation-specific attributes used in Table 2. Panel (B) in the figure shows that the estimated year dummy coefficients are not statistically different from zero for the years of 2001–2004, while those after 2004 are statistically significant and are greater than zero. Given the retroactive nature of the institutional changes and the

TABLE A1. Compliance and Enforcement Over Time

Dependent Variable:	In Compliance (1)	Log (Penalty Amount + 1) (2) (3) (4)		
<i>Year FE</i>				
2001	0.005 (0.020)	0.042 (0.241)	0.047 (0.243)	-0.016 (0.237)
2002	0.025 (0.023)	-0.616 (0.603)	-0.612 (0.572)	-0.643 (0.575)
2003	0.010 (0.025)	-0.348 (0.608)	-0.369 (0.614)	-0.345 (0.610)
2004	0.036 (0.026)	0.262 (0.645)	0.263 (0.646)	0.225 (0.639)
2005	0.018 (0.025)	1.556*** (0.559)	1.558*** (0.555)	1.476*** (0.550)
2006	0.033 (0.026)	1.895*** (0.700)	1.889*** (0.698)	1.824*** (0.678)
2007	0.037 (0.027)	2.426*** (0.615)	2.422*** (0.626)	2.408*** (0.586)
2008	0.053* (0.028)	2.100*** (0.636)	2.106*** (0.638)	2.109*** (0.637)
2009	0.082*** (0.027)	2.957*** (0.597)	2.959*** (0.601)	2.914*** (0.570)
2010	0.047 (0.030)	4.048*** (0.732)	4.051*** (0.725)	4.026*** (0.687)
2011	0.061** (0.031)	3.257*** (1.170)	3.256*** (1.175)	3.274*** (1.122)
2012	0.117*** (0.028)			
2013	0.109*** (0.029)			
2014	0.106*** (0.029)			
<i>Quarter FE</i>				
2 nd quarter (April-June)	0.027*** (0.008)	-0.254 (0.166)	-0.328 (0.212)	-0.235 (0.154)
3 rd quarter (July-September)	0.053*** (0.012)	0.087 (0.121)	-0.003 (0.222)	0.088 (0.119)
4 th quarter (October-December)	0.025*** (0.008)	-0.135 (0.834)	0.171 (0.140)	0.185 (0.137)
<i>(Continued)</i>				

(Continued)

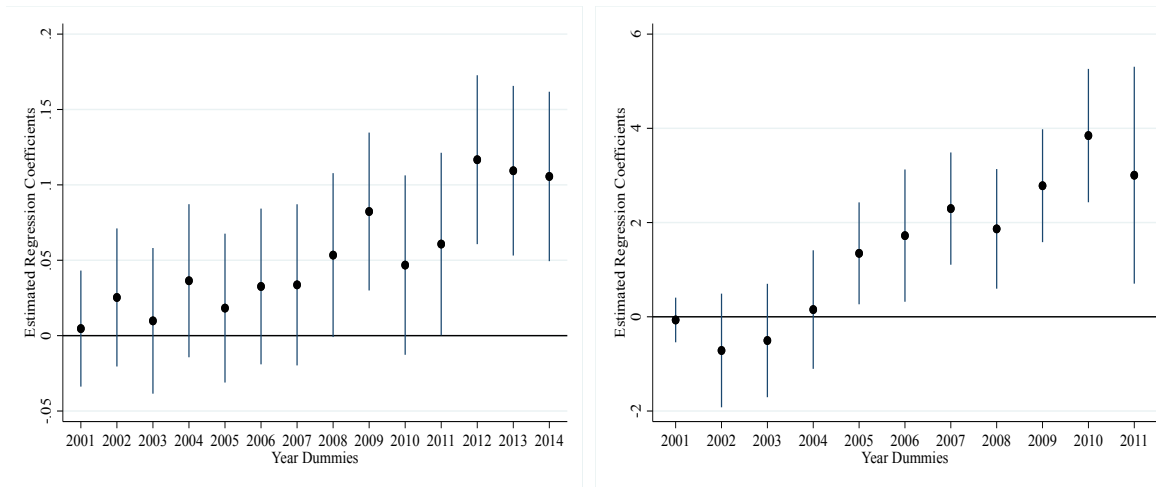
four-year penalty window that we use in this specification, it is not a surprise that there is an upward slopping trend among the estimated year dummy coefficients between 2002 and 2006. These results show that both compliance rates and penalties

TABLE A1. Compliance and Enforcement Over Time (Continued)

Dependent Variable:	In Compliance (1)	Log (Penalty Amount + 1)		
		(2)	(3)	(4)
<i>Weather & pollution</i>				
Precipitation > 75 th percentile	-0.034*** (0.009)	-0.119 (0.175)		
Precipitation < 25 th percentile	0.015** (0.008)	0.019 (0.196)		
Precipitation			-0.012 (0.033)	
Precipitation squared			0.0001 (0.001)	
Precipitation deciles				
10-20 th percentile				-0.263 (0.290)
20-30 th percentile				-0.519 (0.529)
30-40 th percentile				0.100 (0.297)
40-50 th percentile				-0.074 (0.256)
50-60 th percentile				0.129 (0.280)
60-70 th percentile				-0.532* (0.303)
70-80 th percentile				-0.391 (0.298)
80-90 th percentile				-0.252 (0.288)
90-100 th percentile				-0.212 (0.353)
Swimmable	0.019 (0.014)	0.523 (0.396)	0.520 (0.388)	0.534 (0.383)
Facility attributes†	Yes	Yes	Yes	Yes
Violation attributes††	No	Yes	Yes	Yes
Facility FE	Yes	Yes	Yes	Yes
Number of observations	12,779	15,827	15,827	15,827
Adjusted R^2	0.223	0.615	0.615	0.618

Notes: This table reports the OLS regression results where the dependent variables are a dummy variable indicating if the facility was in compliance during the quarterly period (Column (1)) and the logarithm of the sum of one and the amount of penalty imposed within four years of the occurrence date of the violation (Columns (2)–(4)). Standard errors are adjusted for clustering at the facility level, and are provided in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † All facility attributes used in the main structural estimation of the model (see Table 5). †† All violation attributes used in Table 2.

FIGURE A1. Compliance and Enforcement



(A) Probability of Being in Compliance

(B) Average Penalty per MMP Violation

Notes: Panels (A) and (B) show the regression coefficient estimates and the 95% confidence intervals for year dummies, where year 2000 is the base year. The dependent variable in the regression for Panel (A) is the fraction of the domestic wastewater treatment facilities without an effluent MMP violation, and the unit of observations is a facility $\times$ year $\times$ quarter. The dependent variable in the regression for Panel (B) is the average penalty per effluent MMP violation assessed within 4 years of the occurrence of the violation, and the unit of observation is a violation. We control for a variety of facility attributes; see the text for the specifications.

increased after the 2006 institutional changes, even after controlling for an extensive set of (time-varying) facility and local attributes, including treatment technology, capacity, pollution, and weather conditions.

A.2. Static vs. Dynamic Enforcement. We assume that the penalty schedule is static in our analysis, motivated by our findings in Section 3.4. The present section provides additional empirical evidence to support the assumption. Table A2 presents the estimates of a regression model where we assess the extent to which the total amount of penalties for the violations during the quarter is associated with the number of total effluent MMP violations, the number of past effluent MMP violations up to one year, and all facility and local attributes that are described in Table 1. In Specifications (1) and (2) in Table A2, we control for the weighted sum of the number of violations during the past four quarters, where the weights reflect “depreciation” with a rate of 10 percent per quarter, following Blundell et al. (2020). In Specification (3), we control for the number of violations of each of the past four quarters separately. In all three specifications, we do not find that past violations are associated with a

TABLE A2. Does Past Compliance Behavior Matter for Enforcement?

Dependent variable: Log (Penalty Amount + 1)	(1)	(2)	(3)
<i>Current violations</i>			
Number of MMP violations	0.0341*** (0.0122)	0.0358*** (0.0130)	0.0347** (0.0128)
<i>Past violations</i>			
Accumulated number of MMP violations†	-0.0000330 (0.00503)	-0.000264 (0.00578)	
1st lagged number of MMP violations			0.00515 (0.00676)
2nd lagged number of MMP violations			-0.00708 (0.00995)
3rd lagged number of MMP violations			0.00448 (0.00515)
4th lagged number of MMP violations			-0.00392 (0.0124)
Technology/cost/county, weather, pollution attributes	No	Yes	Yes
Regional water board, year, quarter FE's	Yes	Yes	Yes
Adjusted R^2	0.143	0.190	0.190

Notes: This table reports OLS estimates. The unit of observation is a facility-quarter; 2,617 observations. Standard errors are adjusted for clustering at the facility level, and are provided in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † The weighted sum of the number of MMP violations in the past four quarters, where the weight for t^{th} lagged number of violations is 0.9^{t-1} .

higher penalty for a current violation, conditional on the number of violations during the current period and (time-varying) facility and local attributes.

In addition, both the EPA and the California water boards stress the importance of improper maintenance and operation, as opposed to insufficient investment, for explaining violations.¹ We also find evidence corroborating this view by analyzing the description of corrective measures that were planned or taken following the detection of a violation in the data. Out of 3,504 violations with such a description, only 30 percent of them are associated with a need for investment—in the sense that the corrective measure description contains words related to capital investment—while the rest of the violations are associated with short-term measures.²

¹For details, see US Environmental Protection Agency (2004) and California State Water Resources Control Board (2010a). For example, the latter document provides details of a case in which a facility administrator employed uncertified operators and failed to provide adequate supervision to trainees, leading to permit violations.

²The keywords used to classify corrective measure descriptions into capital investment are: *capital*; *construct*; *design*; *fund*; *grant*; *install*; *invest*; *new*; *project*; and *upgrade*.

We argue that the potential efficiency loss associated with the static enforcement scheme as observed in the data, compared to a history-dependent enforcement mechanism, is not likely to be large in our setting. Once initial compliance has been achieved, violations are mainly affected by maintenance and operation of pollution abatement equipment (Harrington, 1988), which is the case for the wastewater treatment facilities. In the engineering literature, for example, Weirich et al. (2015) find that past violations of a wastewater treatment facility increase the likelihood of subsequent violations, but such effect only lasts from 2 to 5 months, depending on capacity and capacity utilization. This contrasts to the other regulatory settings studied by Duflo et al. (2018) and Blundell et al. (2020) where the compliance is mainly achieved by (discrete) investment.

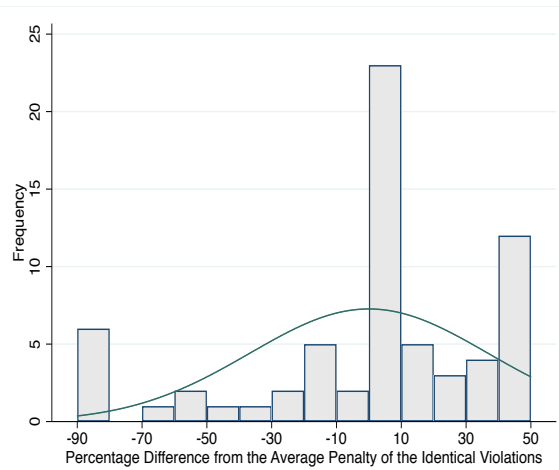
A.3. Discretion in Enforcement.

A.3.1. *Identical Violations.* In Section 3.4, we show that the variation in penalties, conditional on violation priority rank and pollutant category, is explained by the facility and local attributes as well as other violations during the same quarter. This finding does not necessarily mean that regulator's discretion matters, because violation attributes that are not controlled for in the analysis could be driving the conditional variation in penalties observed in the data.

To address this concern, we look at MMP violations that are identical in terms of the pollutant and its permitted and actual amounts of discharge for a given period (e.g. 30-day median, weekly average, etc.). Focusing on the domestic wastewater treatment facilities' violations from 2009–2014 that resulted in a nonzero penalty, we identify 17 unique groups of identical violations by at least three distinct facilities, with a total of 67 violation records. Figure A2 presents a histogram of the percentage difference of the assigned penalty for a violation from the average penalty for its identical violations in the group. The percentage differences range from -84 to 49, and 48 percent of the 67 violations led to penalties that differ by more than 20 percent from the average penalty of their group.

Table A3 shows that this observed dispersion in penalties, conditional on the detailed nature and severity of violation, is explained by the number of concurrent violations during the same quarter of the violation and some of the facility and local attributes. This finding is consistent with our finding from Table 2 in Section 3.4. The specifications of Columns (1) and (2) in Table A3 are similar to Columns (3) and (6) in Table 2—except that, in the latter, we control for violation attributes, while,

FIGURE A2. Heterogeneous Penalties for Identical Violations



Notes: This figure shows a histogram of the percentage difference of the assigned penalty for a given violation from the average penalty for its identical violations in terms of the pollutant and the permitted and actual amounts of discharge, overlaid with a fitted Normal density function. The histogram and the density function are based on 67 violation records with nonzero penalties, grouped into 17 unique cases of identical violations by at least three distinct domestic wastewater treatment facilities during 2009–2014.

in the former, we control for violation fixed effects. These fixed effects are based on pollutant, emission limit and result, and measurement unit and period.³ By controlling for violation fixed effects, we look at how the remaining variation in penalties, as represented in Figure A2, can be explained by the factors considered in Section 3.4. We find that facilities run by a special district, built after 1982, or located in a county with high average household income or a high population density tend to get higher penalties in expectation.

A.3.2. Penalty Assessment: Staff vs. Board. As described in Sections 2.2 and 2.3, the penalty assessed by the enforcement staff at the regional water board can then be adjusted by the board. For a large fraction (86 percent) of the penalty actions of 2009–2015, we have data on the penalty amount assessed by staff as well as the final penalty amount.⁴ Among the 263 penalty actions with the staff assessment

³For Columns (1) and (2) in Table A3, we use all effluent MMP violations by the wastewater treatment facilities in our dataset during 2009–2014. However, the final sample size, 770, is much smaller than that for Table 2 because we control for violation fixed effects, and most violations are unique in terms of pollutant, emission limit and result, and measurement unit and period.

⁴Starting in 2007, it is required to report the staff assessment amount in the data system, but the system may not have been set up to encourage the reporting in the beginning. As a result, we have the staff assessment amount for 11 percent of the penalty actions from 2000–2006, 57 percent for the 2007 penalty actions, and 79 percent for the 2008 penalty actions.

TABLE A3. Further Evidence on Regulatory Discretion

Dependent variable:	Penalized (1)	Log (Penalty+ 1) (2)	Log(Penalty) (3)
Number of current MMP violations	0.00208** (0.000992)	0.0186** (0.00779)	-
Any violations in the past semester	0.0733 (0.0541)	0.505 (0.426)	-
<i>Facility attributes</i>			
Major facility	0.148 (0.201)	1.491 (1.555)	-0.059 (0.101)
First operated in 1982-87	0.298* (0.174)	2.470* (1.337)	-0.109** (0.051)
First operated after 1987	0.326 (0.198)	2.769* (1.507)	-0.117 (0.124)
Advanced or tertiary treatment level	0.114 (0.0953)	0.352 (0.720)	0.035 (0.059)
Capacity utilization > 87%	0.0627 (0.106)	0.496 (0.832)	0.014 (0.061)
Service population < 10,000	-0.119 (0.204)	-0.974 (1.577)	-0.146 (0.091)
Run by a special district	0.299*** (0.0961)	2.060** (0.792)	0.021 (0.060)
High threat to water quality (by the board)	-0.136 (0.113)	-1.229 (0.910)	0.031 (0.056)
<i>County attributes</i>			
Large irrigation water use (> 67%, 2010)	0.200 (0.166)	1.198 (1.322)	-0.171** (0.067)
High income (>\$57K, 2010)	0.296* (0.162)	2.578* (1.303)	0.079 (0.067)
High population density (> 722/mi ² , 2010)	0.201 (0.206)	1.656 (1.662)	-0.066 (0.079)
Low population density (< 80/mi ² , 2010)	-0.527*** (0.134)	-3.404*** (1.030)	0.114 (0.078)
High approval for the 2006 prop. (> 50%)	-0.141 (0.167)	-0.642 (1.336)	-0.004 (0.055)
Log (Staff penalty assessment)	-	-	1.000*** (0.013)
Violation FE†	Yes	Yes	-
Weather and pollution††	Yes	Yes	Yes
Regional water board/year/quarter FEs††	Yes	Yes	Yes
Number of observations	770	770	344
Adjusted R ²	0.675	0.667	0.933

Notes: This table reports OLS estimates. Standard errors are adjusted for clustering at the facility level, and are provided in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The unit of observation is a violation (Columns (1) and (2)) or a penalty action (Column (3)). †: Each fixed effect parameter is associated with a group of violations that share identical pollutant, emission limit and result, and measurement unit and period. ††: All weather and pollution attributes in Table 1. Violation year/quarter fixed effects for (1)–(2); penalty action year fixed effects for (3).

TABLE A4. Number of Permits and Staff Size in FY 2006–2007

Regional Boards	Permits	Compliance Staff	Enforcement Staff	Permits Per Staff
North Coast	1,789	2	3	368
San Francisco Bay	5,116	11	20	165
Central Coast	2,465	11	6	145
Los Angeles	8,078	10	5	539
Central Valley	15,778	21	22	367
Lahontan	2,305	4	6	231
Colorado River	1,636	12	3	109
Santa Ana	7,757	13	8	369
San Diego	5,365	10	4	383

Source: California Water Boards Baseline Enforcement Report, FY 2006–2007. All permits for the 5 core regulatory programs, NPDES, Stormwater, WDRs, Land Disposal, and 401 Certification, are considered in this table. The total regional board staff size is much larger than the sum of compliance and enforcement staff size. For example, there were 113 staff members at the San Francisco Bay Regional Water Quality Control Board during the period.

information, we find that the final penalty amount by the board differs from the initial amount assessed by the staff for 63 penalty actions (24 percent), and is on average 16 percent higher than the initial amount, with the 5th percentile being a decrease of 44 percent and the 95th percentile being an increase of 130 percent.

To investigate the sources of the variation in the adjustment made by the board, we regress the logarithm of the final penalty amount on the logarithm of the initial amount and (time-varying) facility and local attributes. Column (3) of Table A3 provides the regression results. By controlling for the initial penalty amount by the staff, we control for all violation attributes as assessed by the staff, and thus the remaining variation represents the discretion of the regional water board, including any efforts to obtain and evaluate extra, relevant information on violations after the staff decisions. We find that facilities located in a county where more than 2/3 of the fresh water is used for irrigation tend to get a “discount” in penalties, further supporting the idea that regulatory discretion is important in understanding the penalty disparities in the data, consistent with our findings in Section 3.4.

A.4. Heterogenous Enforcement Resources. The regional water boards differ by their resources to enforce the regulations. Table A4 shows the number of permits and staff size by the regional boards.

APPENDIX B. MORE ON THEORY AND IDENTIFICATION

B.1. Proofs for Sections 4 and 5.

B.1.1. *Lemma A1.* Consider any continuous function $e(\cdot)$ with domain $a \in [0, \bar{a}]$, where $\bar{a} < \infty$. Then, for any $\delta > 0$, there exists $\epsilon(\cdot) : \mathbb{N} \rightarrow \mathbb{R}$ such that

$$\sup_{a \in [0, \bar{a}]} \left| e(a) - \exp(-a) \sum_{k=0}^{\infty} \frac{\epsilon(k)}{k!} a^k \right| < \delta.$$

Proof. For any continuous $e(\cdot)$ and $\delta > 0$, by the Weierstrass approximation theorem, the regulator can select a polynomial $\sum_{k=0}^{K_\delta} \beta_k a^k$ such that

$$\sup_{a \in [0, \bar{a}]} \left| e(a) \exp(a) - \sum_{k=0}^{K_\delta} \beta_k a^k \right| < \delta,$$

where we make explicit the dependency between the polynomial degree K_δ and δ (Royden and Fitzpatrick, 1988). With $\epsilon(k) = \beta_k k! \mathbb{1}_{\{k \leq K_\delta\}}$, for every $a \in [0, \bar{a}]$,

$$\left| e(a) - \exp(-a) \sum_{k=0}^{\infty} \frac{\epsilon(k)}{k!} a^k \right| = \exp(-a) \left| e(a) \exp(a) - \sum_{k=0}^{K_\delta} \beta_k a^k \right| < \exp(-a) \delta \leq \delta.$$

□

B.1.2. *Proof of Proposition 1.* Let $\tilde{a}(\theta)$ denote the value of a which satisfies the regulator's first order condition, (9), for a given $\theta \in (0, \bar{\theta})$. In the text, we have shown that the negligence schedule, denoted by $a^*(\theta) \equiv \max\{0, \tilde{a}^*(\theta)\}$ for all θ in its support, is optimal. Given Assumption 1 and our assumption that $F(\cdot)$ is continuously differentiable, $a^*(\cdot)$ is continuous. Here, we show that $\tilde{a}^*(\cdot)$ is strictly increasing, and hence $a^*(\cdot)$ is nondecreasing. Denoting $\varphi(\theta) \equiv (1 - \psi)\theta + \frac{\psi[1-F(\theta)]}{f(\theta)}$ and differentiating both sides of (9) with respect to θ , we obtain:

$$\left(h''[\tilde{a}^*(\theta)] - b''[\tilde{a}^*(\theta)]\varphi(\theta) \right) \tilde{a}'^*(\theta) - b'[\tilde{a}^*(\theta)]\varphi'(\theta) = 0.$$

By Assumption 1, $b'(\cdot) > 0$ and by Assumption 2(i), $\varphi'(\cdot) > 0$. Furthermore, by Assumption 2, $h''[\tilde{a}^*(\theta)] - b''[\tilde{a}^*(\theta)]\varphi(\theta) > 0$, which completes the proof. □

B.1.3. *Proof of Lemma 1.* Fix any time period t . The moment generating function of the number of violations K_t , $M_{K_t}(\cdot)$, is:

$$M_{K_t}(s) = \mathbb{E}_{K_t}(e^{ks}) = \mathbb{E}_{A_t}[\mathbb{E}_{K_t}(e^{ks} | a)] = \mathbb{E}_{A_t}[e^{a(e^s - 1)}] = M_{A_t}(e^s - 1).$$

where the third equality follows from the moment generating function of the Poisson distribution with mean a . Because A_t has a bounded support, $[0, a_t(\bar{\theta})]$, $M_{K_t}(s)$ exists for any $s \in \mathbb{R}$. Letting $u = e^s - 1$ shows $M_{A_t}(u) = M_{K_t}[\log(1 + u)]$, for $u \in (-1, \infty)$. Hence, $M_{A_t}(\cdot)$ is identified on a neighborhood of 0, thereby identifying $G_t(\cdot)$. $\square$

B.1.4. *Proof of Lemma 2.* The first equation follows from $F(\cdot)$ and $b(\cdot)$ not changing over time and from the strict monotonicity $\tilde{a}(\cdot, j)$ in its first argument. Concerning the second equation, from (3), we have that $\tilde{\theta}(a, j)b'(a) = e'_j(a)$ for $j \in \{pre, post\}$, which implies that $\tilde{\theta}(a, post) = \frac{e'_{post}(a)}{e'_{pre}(a)}\tilde{\theta}(a, pre)$. $\square$

B.1.5. *Proof of Proposition 2.* We first show by induction that $\theta_l = \tilde{\theta}(a_l, post) = \tilde{\theta}(a_{l+1}, pre)$. From the normalization, $\theta_0 = \tilde{\theta}(a_0, post)$. For any l , let $\theta_l = \tilde{\theta}(a_l, post)$. Then, $a_{l+1} = T^H(a_l) = \tilde{a}[\tilde{\theta}(a_l, post), pre] = \tilde{a}(\theta_l, pre)$, where the first and second equalities are due to the definition of a_{t+1} and Lemma 2, respectively. Thus, $\theta_l = \tilde{\theta}(a_{l+1}, pre)$. Moreover, $\theta_{l+1} = T^V(\theta_l, a_{l+1}) = T^V[\tilde{\theta}(a_{l+1}, pre), a_{l+1}]$, where the second equality is due to the definition of θ_{l+1} . Therefore, from Lemma 2, we have that $\theta_{l+1} = \tilde{\theta}(a_{l+1}, post)$. We can then use (3) to write $b'(a_l) = \frac{e'_{pre}(a_l)}{\theta_{l-1}} = \frac{e'_{post}(a_l)}{\theta_l}$. Moreover, $F(\theta_l)$ is identified by $F(\theta_l) = G_{post}(a_l) = G_{pre}(a_{l+1})$. $\square$

B.1.6. *Proof of Proposition 3.* Let $Q(\alpha)$ denote α -quantile of $F(\cdot)$. We can rewrite equation (9) as

$$b'[G_j^{-1}(\alpha)] \left[(1 - \psi_j)Q(\alpha) + \frac{\psi_j(1 - \alpha)}{f[Q(\alpha)]} \right] = \sum_{r=1}^R \gamma_{j,r} [G_j^{-1}(\alpha)]^{r-1}. \quad (\text{A.1})$$

We may also rewrite equation (3) as:

$$e'_j[G_j^{-1}(\alpha)] = Q(\alpha)b'[G_j^{-1}(\alpha)]. \quad (\text{A.2})$$

Using equation (A.2) and the relationship between the density and its quantile function, i.e., $f[Q(\alpha)] = 1/Q'(\alpha)$, we rewrite equation (A.1) as

$$\frac{e'_j[G_j^{-1}(\alpha)]}{Q(\alpha)} [(1 - \psi_j)Q(\alpha) + Q'(\alpha)\psi_j(1 - \alpha)] = \sum_{r=1}^R \gamma_{j,r} [G_j^{-1}(\alpha)]^{r-1},$$

which implies

$$\frac{Q'(\alpha)}{Q(\alpha)} = \frac{\sum_{r=1}^R \gamma_{j,r} [G_j^{-1}(\alpha)]^{r-1} - e'_j[G_j^{-1}(\alpha)](1 - \psi_j)}{e'_j[G_j^{-1}(\alpha)]\psi_j(1 - \alpha)}, \quad (\text{A.3})$$

for $j \in \{pre, post\}$. Define $\Gamma_{j,r} \equiv \frac{\gamma_{j,r}}{\psi_j}$ and $\Psi_j = \frac{1-\psi_j}{\psi_j}$, and notice that there is a one-to-one relationship between $\left(\{\Gamma_{j,r}\}_{r=1}^R, \Psi_j\right)$ and $\left(\{\gamma_{j,r}\}_{r=1}^R, \psi_j\right)$. Integrating the above equation from some α_0 to α gives

$$\log \frac{Q(\alpha)}{Q(\alpha_0)} = \int_{\alpha_0}^{\alpha} \left(\sum_{r=1}^R \Gamma_{j,r} \frac{[G_j^{-1}(u)]^{r-1}}{e'_j[G_j^{-1}(u)]} - \Psi_j \right) \frac{1}{(1-u)} du. \quad (\text{A.4})$$

Remember that $F(\theta_l) = G_j[\tilde{a}(\theta_l, j)]$. From Proposition 2, there is a vector $\{\theta_l\}_{l=0}^{\bar{L}}$ such that $\tilde{a}(\theta_l, j)$ is known for $j \in \{pre, post\}$. Since equation (A.4) holds for arbitrary α and α_0 , the following holds for any $l \in \{1, \dots, \bar{L}\}$ and $j \in \{pre, post\}$:

$$\log \frac{\theta_l}{\theta_0} = \sum_{r=1}^R \Gamma_{j,r} \int_{G_j[\tilde{a}(\theta_0, j)]}^{G_j[\tilde{a}(\theta_l, j)]} \frac{[G_j^{-1}(u)]^{r-1}}{e'_j[G_j^{-1}(u)](1-u)} du - \Psi_j \int_{G_j[\tilde{a}(\theta_0, j)]}^{G_j[\tilde{a}(\theta_l, j)]} \frac{1}{(1-u)} du. \quad (\text{A.5})$$

Furthermore, we obtain the following equations for any α by observing that equation (A.3) holds for both regimes:

$$\frac{\sum_{r=1}^R \Gamma_{post,r} [G_{post}^{-1}(\alpha)]^{r-1}}{e'_{post}[G_{post}^{-1}(\alpha)]} - \frac{\sum_{r=1}^R \Gamma_{pre,r} [G_{pre}^{-1}(\alpha)]^{r-1}}{e'_{pre}[G_{pre}^{-1}(\alpha)]} + \Psi_{pre} - \Psi_{post} = 0. \quad (\text{A.6})$$

For each regime, equation (A.5) specifies a system of $\bar{L}$ linear equations and $R + 1$ unknowns $\left(\{\Gamma_{j,r}\}_{r=1}^R \text{ and } \Psi_j\right)$, and equation (A.6) specifies an infinite number of equations. Assumption 5 suffices for a system consisting of equations (A.5) and (A.6) to have a unique solution for $\{\gamma_{pre,r}\}_{r=1}^R, \{\gamma_{post,r}\}_{r=1}^R, \psi_{pre}$ and ψ_{post} . Now, setting $\alpha_0 = G_j(a_0)$ in equation (A.4), we identify $Q(\cdot)$ and, accordingly, $F(\cdot)$ and $f(\cdot)$. Lastly, using equation (A.1), we identify $b'(a)$ for $a \in \mathcal{A}_{pre} \cup \mathcal{A}_{post}$.

Our model is over-identified because we can evaluate (A.6) at an arbitrarily large number of quantiles. Moreover, for each regime, there is at least one more equation that we could use for the identification of the model primitives, which we obtain by evaluating equation (9) at the upper bounds of $\mathcal{A}_j$'s. $\square$

B.2. Comparative Statics. We study how the equilibrium violations and the stringency in enforcement vary with the regulator preferences (γ, ψ) and the compliance costs. To simply capture the heterogeneity of costs, we introduce a parameter, β , which enters in $b'(a, \beta)$. We assume that $\partial b'(a, \beta) / \partial \beta > 0$.

COROLLARY A1. *Under Assumptions 1 and 2, for any $\theta \in (0, \bar{\theta})$, (i) the optimal negligence level for θ , $a(\theta)$, is decreasing in γ ; (ii) if $-\theta + \frac{1-F(\theta)}{f(\theta)} > 0$ and $b''(\cdot) < 0$, then $a(\theta)$ is increasing in ψ ; and (iii) if $b''(\cdot) < 0$, $a(\theta)$ is increasing in β .*

Proof. The comparative statics on violations are based on the regulator's first order condition:

$$\gamma - b'[a(\theta)] \left((1 - \psi)\theta + \frac{\psi\{1 - F(\theta)\}}{f(\theta)} \right) = 0. \quad (\text{A.7})$$

First, taking a derivative of (A.7) with respect to γ :

$$1 - b''[a(\theta)] \left((1 - \psi)\theta + \frac{\psi\{1 - F(\theta)\}}{f(\theta)} \right) \frac{\partial a(\theta)}{\partial \gamma} = 0.$$

Rearranging the above equation:

$$\frac{\partial a(\theta)}{\partial \gamma} = \left[b''[a(\theta)] \left((1 - \psi)\theta + \frac{\psi\{1 - F(\theta)\}}{f(\theta)} \right) \right]^{-1}.$$

By Assumption 2, $a(\theta)$ is decreasing in γ for any θ . Second, taking a derivative of (A.7) with respect to ψ :

$$-b''[a(\theta)] \left((1 - \psi)\theta + \frac{\psi\{1 - F(\theta)\}}{f(\theta)} \right) \frac{\partial a(\theta)}{\partial \psi} - b'[a(\theta)] \left(-\theta + \frac{1 - F(\theta)}{f(\theta)} \right) = 0.$$

Rearranging the above equation:

$$\frac{\partial a(\theta)}{\partial \psi} = -b'[a(\theta)] \left(-\theta + \frac{1 - F(\theta)}{f(\theta)} \right) \left[b''[a(\theta)] \left((1 - \psi)\theta + \frac{\psi\{1 - F(\theta)\}}{f(\theta)} \right) \right]^{-1}.$$

Under Assumptions 1 and 2, $a(\theta)$ is increasing in ψ if $\left(-\theta + \frac{1 - F(\theta)}{f(\theta)} \right)$ is positive for any given θ . Lastly, taking a derivative of (A.7) with respect to β :

$$- \left((1 - \psi)\theta + \frac{\psi\{1 - F(\theta)\}}{f(\theta)} \right) \left(\frac{\partial b'[a(\theta)]}{\partial a} \frac{\partial a(\theta)}{\partial \beta} + \frac{\partial b'[a(\theta)]}{\partial \beta} \right) = 0.$$

Rearranging the above equation:

$$\frac{\partial a(\theta)}{\partial \beta} = -\frac{1}{b''[a(\theta)]} \left(\frac{\partial b'[a(\theta)]}{\partial \beta} \right).$$

Thus, as $b(\cdot)$ is strictly concave, $a(\theta)$ is increasing in β for any θ . $\square$

We consider the stringency of enforcement as $e'[a(\theta)]$. The following proposition shows our comparative statics results on $e'[a(\theta)]$.

COROLLARY A2. *Suppose the assumptions for Proposition 1 are satisfied. Then, for any $\theta \in (0, \bar{\theta})$, (i) $e'[a(\theta)]$ is increasing in γ ; (ii) if $-\theta + \frac{1 - F(\theta)}{f(\theta)} > 0$ and $b''(\cdot) < 0$, then $e'[a(\theta)]$ is decreasing in ψ ; and (iii) if $b''(\cdot) < 0$, then $e'[a(\theta)]$ is decreasing in β .*

Proof. We start with the facility's first order condition:

$$\theta b'[a(\theta)] = e'[a(\theta)], \quad (\text{A.8})$$

First, taking a derivative of (A.8) with respect to γ :

$$\theta b''[a(\theta)] \frac{\partial a(\theta)}{\partial \gamma} = \frac{\partial e'[a(\theta)]}{\partial \gamma} + e''[a(\theta)] \frac{\partial a(\theta)}{\partial \gamma}.$$

We have shown that $a(\theta)$ is decreasing in γ and $a(\cdot)$ is implemented. The latter implies that the second order condition for the facility holds. Hence, $e'[a(\theta)]$ is increasing in γ for any θ . Second, taking a derivative of (A.8) with respect to ψ :

$$\theta b''[a(\theta)] \frac{\partial a(\theta)}{\partial \psi} = \frac{\partial e'[a(\theta)]}{\partial \psi} + e''[a(\theta)] \frac{\partial a(\theta)}{\partial \psi}.$$

Because the second order condition for the facility holds, $e'[a(\theta)]$ is decreasing in ψ if $a(\theta)$ is increasing in ψ . Lastly, taking a derivative of (A.8) with respect to β :

$$\theta b''[a(\theta)] \frac{\partial a(\theta)}{\partial \beta} + \theta \frac{\partial b'[a(\theta)]}{\partial \beta} = \frac{\partial e'[a(\theta)]}{\partial \beta} + e''[a(\theta)] \frac{\partial a(\theta)}{\partial \beta}.$$

Because the second order condition for the facility holds, $e'[a(\theta)]$ is increasing in β for any given θ if $a(\theta)$ is increasing in β . $\square$

APPENDIX C. ESTIMATION PROCEDURE

Step 1. We parametrically estimate the expected penalties as specified in (13) by MLE. Given the estimates, we estimate the marginal expected penalty, $\hat{e}'_j(a|\mathbf{x})$ using equation (1) for $j = \{pre, post\}$. To estimate $G_j(\cdot|\mathbf{x})$, we use the parametric specification presented in Section 5.3.2. Such a specification implies that the number of violations follows a Poisson-Gamma distribution, which is equivalent to a negative binomial distribution with mean $\exp(\beta_{0,j} + \beta_1 \mathbf{x}_{i,t})$ and variance $\exp(\beta_{0,j} + \beta_1 \mathbf{x}_{i,t}) [1 + \Delta(\mathbf{z}_{i,t})^{-1} \exp(\beta_{0,j} + \beta_1 \mathbf{x}_{i,t})]$, where $\Delta(\mathbf{z}_{i,t}) \equiv \exp(\mathbf{z}_{i,t} \delta)$ is the overdispersion parameter. We are thus able to estimate δ and β_j 's by MLE, using a negative binomial regression. See Cameron and Trivedi (2013) for details about this estimator.

Step 2. We denote by $\hat{\theta}(a, j|\mathbf{x})$ an estimator of the facility type that sets negligence level a under regime j , given $\mathbf{x}$. We normalize $\hat{\theta}(1, post) = 1$, and employ the empirical counterparts of the transforms T^H and T^V , defined in (11) and (12), to obtain $\hat{\theta}(a, j|\mathbf{x})$

for a sequence of values of a . Normalizing $\hat{a}_0(\mathbf{x}) = \hat{\theta}_0(\mathbf{x}) = 1$, we define recursively:

$$\hat{a}_l(\mathbf{x}) \equiv \hat{G}_{pre}^{-1} \left[\hat{G}_{post} [\hat{a}_{l-1}(\mathbf{x}) | \mathbf{x}] | \mathbf{x} \right],$$

$$\text{and} \quad \hat{\theta}_l(\mathbf{x}) \equiv \frac{\hat{e}'_{post} [\hat{a}_l(\mathbf{x}) | \mathbf{x}]}{\hat{e}'_{pre} [\hat{a}_l(\mathbf{x}) | \mathbf{x}]} \hat{\theta}_{l-1}(\mathbf{x}).$$

Define $\hat{\theta}_l^{post}(\mathbf{x}) \equiv \hat{\theta}_l(\mathbf{x})$, $\hat{\theta}_l^{pre}(\mathbf{x}) \equiv \hat{\theta}_{l-1}(\mathbf{x})$, $\hat{a}_l^{post}(\mathbf{x}) \equiv \hat{a}_l(\mathbf{x})$ and $\hat{a}_l^{pre}(\mathbf{x}) \equiv \hat{a}_l(\mathbf{x})$, for every l . We use $\hat{\theta}_l^j(\mathbf{x})$ as an estimator of $\tilde{\theta}(\hat{a}_l^j(\mathbf{x}), j | \mathbf{x})$, for $j \in \{pre, post\}$ and any l .

Step 3. Equation (A.5) implies that

$$\sum_l \left\{ \log \frac{\theta_l}{\theta_0} - \sum_{r=1}^R \Gamma_{j,r} \int_{\alpha_0}^{\alpha_l} \frac{[G_j^{-1}(u)]^{r-1}}{e'_j [G_j^{-1}(u)] (1-u)} du + \Psi_j \int_{\alpha_0}^{\alpha_l} \frac{1}{(1-u)} du \right\}^2 = 0,$$

for $j \in \{pre, post\}$. Also, from (A.6), we have

$$\sum_{\alpha \in U} \left\{ \frac{\sum_{r=1}^R \Gamma_{post,r} [G_{post}^{-1}(\alpha)]^{r-1}}{e'_{post} [G_{post}^{-1}(\alpha)]} + \Psi_{pre} - \frac{\sum_{r=1}^R \Gamma_{pre,r} [G_{pre}^{-1}(\alpha)]^{r-1}}{e'_{pre} [G_{pre}^{-1}(\alpha)]} - \Psi_{post} \right\}^2 = 0,$$

where $U = \{\alpha_1, \dots, \alpha_{N_U}\}$ is a grid in the $(0, 1)$ interval such that $G_{post}^{-1}(\alpha) > 0$ for all $\alpha \in U$. We estimate $\{\Gamma_{pre,r}(\mathbf{x})\}_{r=1}^R$, $\Psi_{pre}(\mathbf{x})$, $\{\Gamma_{post,r}(\mathbf{x})\}_{r=1}^R$ and $\Psi_{post}(\mathbf{x})$ using a sample analogue of the above two equations for any given $\mathbf{x}$. In our application, the system comprised by these equations is over-identified, so we employ the least-squares solution. We then estimate $\{\gamma_{j,r}(\mathbf{x})\}_{r=1}^R$ and $\psi_j(\mathbf{x})$ as

$$\hat{\psi}_j(\mathbf{x}) \equiv \frac{1}{1 - \hat{\Psi}_j(\mathbf{x})} \quad \text{and} \quad \hat{\gamma}_{j,r}(\mathbf{x}) \equiv \hat{\Gamma}_{j,r}(\mathbf{x}) \hat{\psi}_j(\mathbf{x}),$$

for $j \in \{pre, post\}$ and $r = \{1, \dots, R\}$. In solving the equations, we constrain the least-squares solution to ensure all regulator parameter estimates are positive.⁵

Step 4. From the empirical analogue to (A.4) with $\alpha_0 = 0$, we estimate the quantile function associated with the distribution of types, conditional on $\mathbf{x}$, as

$$\hat{Q}_j(\alpha | \mathbf{x}) \equiv \hat{\theta}_0^j(\mathbf{x}) \exp \left(\int_{\hat{G}_j[\hat{a}_0^j(\mathbf{x}) | \mathbf{x}]}^{\alpha} \left[\sum_{r=1}^R \Gamma_{j,r}(\mathbf{x}) \frac{[\hat{G}_j^{-1}(u | \mathbf{x})]^{r-1}}{e'_j [\hat{G}_j^{-1}(u | \mathbf{x}) | \mathbf{x}]} - \Psi_j(\mathbf{x}) \right] \frac{1}{(1-u)} du \right),$$

⁵We find that the constrained estimates are different from the unconstrained ones for 25 (11 percent) out of the 221 facilities in estimation.

for $j \in \{pre, post\}$. An estimator for $F(\cdot|\mathbf{x})$, or $\hat{F}_j(\cdot|\mathbf{x})$, is the inverse of $\hat{Q}_j(\cdot|\mathbf{x})$ —which, under Assumption 2, is guaranteed to exist. Finally, we define

$$\hat{b}'_j(a|\mathbf{x}) \equiv \hat{e}'_j(a|\mathbf{x}) / \hat{Q}_j[\hat{G}_j(a|\mathbf{x})|\mathbf{x}],$$

and numerically integrate the above expression over a to obtain $\hat{b}_j(a|\mathbf{x})$. For any finite sample, we might have $\hat{F}_{pre}(\cdot|\mathbf{x}) \neq \hat{F}_{post}(\cdot|\mathbf{x})$ and $\hat{b}'_{pre}(\cdot|\mathbf{x}) \neq \hat{b}'_{post}(\cdot|\mathbf{x})$, even though our estimator is consistent. In our counterfactual analyses, we employ $\hat{F}_{pre}(\cdot|\mathbf{x})$ and $\hat{b}'_{pre}(\cdot|\mathbf{x})$ for scenarios referring to the pre-2006 period, and $\hat{F}_{post}(\cdot|\mathbf{x})$ and $\hat{b}'_{post}(\cdot|\mathbf{x})$ for post-2006 scenarios.

APPENDIX D. COUNTERFACTUAL ANALYSES DETAILS

We now provide details on the implementation of the counterfactual analyses described in Section 6.5. As in our empirical application, we assume that the social costs of negligence are linear—that is, $R = 1$.

D.1. Uniform Penalties. We start by considering the case in which the regulator chooses a single nonlinear penalty schedule to be applied to all facilities. For computational convenience, we restrict our attention to polynomial schedules of degree three. Specifically we assume that the regulator sets $\tilde{\epsilon} \equiv \{\epsilon_1, \epsilon_2, \epsilon_3\}$, and the penalty for a facility with k violations in a quarter is

$$\epsilon(k; \tilde{\epsilon}) = \begin{cases} \epsilon_1 k + \epsilon_2 k^2 + \epsilon_3 k^3 & \text{if } k \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Given our assumption that the number of violations follows a Poisson distribution with mean a , we can then write the expected penalty as

$$e(a; \tilde{\epsilon}) = \mathbb{E}[\epsilon(k; \tilde{\epsilon})|a] = \epsilon_1 a + \epsilon_2(a + a^2) + \epsilon_3(a + 3a^2 + a^3),$$

for a facility that sets the negligence level $a \geq 0$.⁶ From the facility's FOC (3), and given a choice of $\tilde{\epsilon}$ by the regulator, we compute the negligence level set by a facility with type θ and observed attributes $\mathbf{x}$ as the solution to

$$\theta \hat{b}'(a|\mathbf{x}) = \epsilon_1 + \epsilon_2(1 + 2a) + \epsilon_3(1 + 6a + 3a^2).$$

⁶This equation is based on the known closed form solution for the following power series: (i) $\sum_{k=0}^{\infty} \frac{a^k}{k!} = e^a$; (ii) $\sum_{k=0}^{\infty} k \frac{a^k}{k!} = ae^a$; (iii) $\sum_{k=0}^{\infty} k^2 \frac{a^k}{k!} = (a + a^2)e^a$; and (iv) $\sum_{k=0}^{\infty} k^3 \frac{a^k}{k!} = (a + 3a^2 + a^3)e^a$.

Let $a(\theta, \mathbf{x}, \tilde{\epsilon})$ denote this solution. We can then take the expectation of (4) over the facility attributes $\mathbf{X}$ to write the objective function of the regulator as

$$\mathbb{E}_{\mathbf{X}} \left(\mathbb{E}_{\Theta|\mathbf{X}} \left[-\theta \hat{b}(a(\theta, \mathbf{x}, \tilde{\epsilon})|\mathbf{x}) + \hat{\gamma}_j(\mathbf{x})a(\theta, \mathbf{x}, \tilde{\epsilon}) + \hat{\psi}_j(\mathbf{x})e(a(\theta, \mathbf{x}, \tilde{\epsilon}); \tilde{\epsilon}) \mid \mathbf{x} \right] \right),$$

$j \in \{pre, post\}$. We minimize this objective function numerically to obtain $\epsilon_1 = 1,764.10$, $\epsilon_2 = 0.00$ and $\epsilon_3 = 2.47$ for $j = pre$; and $\epsilon_1 = 2,360.98$, $\epsilon_2 = 45.94$ and $\epsilon_3 = -0.48$ for $j = post$.

In the main text we report that the uniform schedule above leads to a reduction in negligence levels of six percent, but it also increases the total amount of penalties, relative to the baseline enforcement schedule. To obtain a flexible schedule that achieves the same reduction in negligence levels without increasing the assigned penalties, we proceed as follows: first we classify each facility as having either high or low compliance costs, based on whether their median marginal cost, defined in footnote 39, is above or below the median marginal cost across all facilities. Then, given an arbitrary number ξ_H , and for every facility with high compliance cost, we use the FOC (3) to determine the penalty schedule that would induce an increase of ξ_H percent in the negligence level set by every possible realization of that facility's type Θ , relative to the baseline scenario. Similarly, given an arbitrary ξ_L , and for every facility with low compliance cost, we determine the penalty schedule that would lead to a decrease of ξ_L percent in the negligence level set by every possible realization of that facility's type. In other words, starting from the baseline scenario, we manipulate the penalty schedules so that high compliance cost facilities are expected to violate more, whereas low cost facilities are expected to violate less. By setting $\xi_H = 20$ and $\xi_L = 30.5$, we obtain a reduction of six percent in the average violation frequency across all facilities—exactly the same reduction that would arise under the uniform policy. Moreover, the total amount of penalties under this targeted manipulation of the schedules would be 2.5 percent below that in the uniform scenario. We would like to stress that the manipulation defined above is just one example, and a particularly simple one. It is likely that another choice of the values ξ_H and ξ_L or a partition of the facilities into more than two compliance cost groups would achieve even larger reductions in the total penalties, relative to the uniform scenario, while maintaining the same average violation frequency.

D.2. Linear Penalties. Now we address the computation of the optimal linear penalties, from the regulator's perspective. Consider a facility with observed attributes $\mathbf{x}$, and let $\hat{b}'^{-1}(\cdot|\mathbf{x})$ denote the inverse of $\hat{b}'(\cdot|\mathbf{x})$. From the FOC of the facility,

(3), given per-violation penalty p , the facility chooses its negligence level according to $a = \hat{b}'^{-1}(p/\theta|\mathbf{x})$. Thus we may rewrite (4), the regulator's objective function, as

$$\mathbb{E}_{\Theta|\mathbf{x}} \left[-\theta \hat{b} \left(\hat{b}'^{-1}(p/\theta|\mathbf{x}) \mid \mathbf{x} \right) + \hat{b}'^{-1}(p/\theta|\mathbf{x}) \left(\hat{\gamma}_j(\mathbf{x}) + \hat{\psi}_j(\mathbf{x})p \right) \mid \mathbf{x} \right],$$

subject to $p > 0$, for $j \in \{pre, post\}$. We minimize this function numerically.

Finally we consider the computation of the linear penalty that leads to the same expected number of violations as in the baseline scenario. Let $\hat{a}_j(\mathbf{x})$ denote the mean baseline negligence level set by a facility with observed attributes $\mathbf{x}$ in period $j \in \{pre, post\}$. We numerically solve for p the equation

$$\mathbb{E}_{\Theta|\mathbf{x}} \left[\hat{b}'^{-1} \left(p/\theta \mid \mathbf{x} \right) \mid \mathbf{x} \right] - \hat{a}_j(\mathbf{x}) = 0,$$

using the optimal linear schedule calculated above as a starting point.

APPENDIX E. FURTHER DISCUSSIONS ON THE ESTIMATION RESULTS

This section presents empirical results that, due to space considerations, we do not show in the main text. First, we present the results from the first step in the estimation procedure. Second, we argue that the estimated model satisfies the assumptions in Sections 4 and 5, which are sufficient for equilibrium characterization and the identification of model primitives. Third, we present the distribution of the regulator preference estimates by the regional board. Fourth, we report our results based on the estimated regulator preferences in the post-2006 period. Fifth, we provide further results on the extent to which the heterogeneity in regulator preferences explain penalty disparities across facilities. Sixth, we discuss how to compute the regulator's expected costs based on our estimates and how these costs vary across the baseline scenario and each of the five counterfactual scenarios of Table 6 in Section 6. Lastly, we provide the results of five additional counterfactual analyses, which we refer to throughout the main text but do not include in Table 6.

E.1. Penalty Schedule and Negligence Distribution Estimates. In Table A5, we present the penalty schedule estimates, ϕ 's in (13), and the estimates of the negligence distributions, δ and β 's in Section 5.3.2. Consistent with the preliminary results discussed in Section 3, we find that $\hat{\phi}_{1,post} > 0$. That is, given the same number of violations, a facility expects to pay a higher penalty in the period following the changes, relative to the prior periods. Furthermore, $\hat{\beta}_{0,pre} - \hat{\beta}_{0,post} < 0$; the facilities decrease negligence levels after the changes.

TABLE A5. Penalty Schedule and Negligence Distribution Estimates

	Penalties		Negligence	
$\phi_{1,post}$	0.76*** (0.15)		-	
$\phi_{1,k}$	0.05*** (0.01)		-	
ϕ_{2,k^2}	4.94 (9.18)		-	
σ_2	0.50*** (0.06)		-	
σ_{12}	0.27* (0.16)		-	
$\beta_{0,post} - \beta_{0,pre}$	-		-0.26*** (0.10)	
	$\phi_{1,x}$	$\phi_{2,x}$	β_1	δ
<i>Facility attributes</i>				
Major facility	0.49*** (0.21)	0.12 (0.10)	-0.20 (0.14)	-
First operated in 1982-87	-0.04 (0.18)	-0.01 (0.08)	-0.06 (0.15)	-
First operated after 1987	-1.28*** (0.56)	-0.01 (0.17)	-0.35 (0.27)	-
Advanced or tertiary	0.06 (0.15)	0.05 (0.07)	-0.22* (0.12)	-
Capacity utilization > 87%	0.19 (0.16)	-0.01 (0.09)	-0.24** (0.11)	-
Service pop. < 10K	0.20 (0.21)	-0.07 (0.07)	0.12 (0.15)	-
Run by a special district	-0.13 (0.15)	-0.14** (0.07)	-0.38*** (0.12)	-
High threat to water	0.24 (0.16)	0.08 (0.06)	-0.28 (0.13)	-
<i>Weather & pollution</i>				
Precipitation > 75 th pct.	0.04 (0.17)	0.00 (0.06)	-0.03 (0.12)	-
Precipitation < 25 th pct.	0.29 (0.23)	0.04 (0.11)	-0.06 (0.13)	-
Swimmable	-0.13 (0.14)	0.00 (0.06)	-0.14 (0.11)	-
<i>County attributes</i>				
Irrigation use > 67%	-0.15 (0.18)	-0.05 (0.10)	0.34** (0.15)	-
Household income > \$57K	0.50** (0.24)	0.25 (0.16)	0.40 (0.17)	-
Pop. density > 722/mi ²	-0.36 (0.27)	-0.05 (0.13)	-0.51** (0.19)	-
Pop. density < 80/mi ²	-0.12 (0.21)	0.30*** (0.12)	-0.29 (0.19)	-
Prop. approval > 50%	0.28 (0.25)	0.16 (0.11)	-0.43** (0.19)	-
<i>Regional water board FE†</i>				
Region 2: San Francisco Bay	1.62*** (0.37)	0.09 (0.18)	-1.31*** (0.25)	-0.84*** (0.18)
Region 3: Central Coast	0.82 (0.38)	-0.05 (0.19)	-0.83*** (0.28)	-0.41* (0.21)
Region 4: Los Angeles	-0.08 (0.37)	-0.40** (0.16)	0.78 (0.26)	-0.73 (0.16)
Region 5: Central Valley	0.99 (0.26)	0.19 (0.18)	0.21 (0.18)	-0.43*** (0.12)
Region 7: Colorado River	1.17*** (0.34)	0.33 (0.20)	0.42 (0.21)	-0.34 (0.17)
Region 8: Santa Ana	2.32* (1.19)	-0.09 (0.30)	0.29 (0.27)	-0.95*** (0.17)
Region 9: San Diego	1.79 (5.24)	0.08 (0.24)	-0.09 (0.98)	-2.93*** (0.41)
Constant	-1.63*** (0.48)	7.51*** (0.27)	0.94*** (0.36)	-1.89*** (0.10)
Number of observations	729		8,429	

Notes: Bootstrap standard errors are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. We estimate the parameters of the penalty schedules in (13) by MLE, employing facility-quarter observations of 2000-2001 and 2009-2010. The parameters of the negligence distributions are estimated by MLE, based on facility-quarter observations in the periods 2002-2005 and 2011-2014. See Section 3.1 and Table 1 for the description of the control variables. † Region 1 (North Coast) is omitted.

E.2. Model Primitives and the Assumptions. Assumptions 1 and 2 in Section 4 are used in Proposition 1 to characterize the equilibrium negligence schedule. For every facility, the estimated γ_{pre} and γ_{post} are positive, and the estimated $b'(\cdot)$ is positive all over its domain. Thus, Assumption 1 is satisfied. As for Assumption 2(i), our estimated distribution of types for each facility satisfies the strict monotonicity of $\varphi_j(\theta) \equiv (1 - \psi_j)\theta + \frac{\psi_j[1-F(\theta)]}{f(\theta)}$ in θ , for $j \in \{pre, post\}$. In addition, given that the estimates of $\varphi_j(\theta)$ are positive for all θ and $j \in \{pre, post\}$, $\varphi_j(\cdot) > 0$. Our estimates also satisfy Assumption 2(ii): the estimated function $b'(\cdot)$ is strictly decreasing for all facilities; and $h(\cdot)$ is assumed to be linear.

Assumption 3 is used in Proposition 2 for the partial identification of the distribution of compliance costs. In our estimation of the enforcement function specified in (13), the parameter $\phi_{1,post}$ is positive, implying that the marginal enforcement function in the post-2006 period is always above that in the pre-2006 period.

We employ Assumptions 4 and 5 in Proposition 3 to achieve the identification of the regulator preferences and the exact identification of the facilities' compliance cost distribution. Since we do not directly observe the function $h(\cdot)$, Assumption 4 is non-testable. That we are able to solve the system of equations implied by (A.5) and (A.6) indicates that our estimates satisfy Assumption 5.

E.3. Distribution of the Estimated Regulator Preferences. Table A6 shows the distribution of the estimated regulator preference parameters for each regional board. There is a considerable level of heterogeneity across regional boards: prior to the 2006 institutional change, the average γ estimates within a regional board varies from 2,490 (Region 1) to 5,486 (Region 7), and the average ψ estimates varies from 0.40 (Region 9) to 1.18 (Region 1). The within-region heterogeneity in the γ and ψ estimates is also large, reflecting variation in observed facility attributes other than the regional boards, such as the facility size and the residents' average income in the facility location (Table 5 in Section 6.3); for example, the γ estimates prior to 2006 in Region 4 range from 1,066 to 7,878.

E.4. Results based on the Regulator Preferences after the 2006 Changes. Most results presented in Section 6 are based on the model primitive estimates for the periods prior to the 2006 institutional changes. Here we provide the results based on the estimates for the periods after the 2006 changes. Given that these results are qualitatively similar to those presented in Section 6, we omit the bootstrap confidence intervals. Table A7 documents the results of regressing the logarithm of

TABLE A6. Distribution of the Estimated Regulator Preferences

	Before the 2006 Changes				After the 2006 Changes			
	Mean	SD	Min	Max	Mean	SD	Min	Max
<i>Marginal external costs (γ), by regional water boards</i>								
Region 1: North Coast	2,490	1,025	898	4,183	2,653	775	1,378	3,980
Region 2: San Francisco Bay	3,478	371	2,745	4,050	3,510	394	2,811	4,143
Region 3: Central Coast	3,170	598	2,241	4,516	2,935	474	2,154	3,722
Region 4: Los Angeles	2,955	1,977	1,066	7,878	2,197	618	1,406	3,091
Region 5: Central Valley	4,658	995	2,734	8,020	3,434	570	2,291	4,782
Region 7: Colorado River	5,486	880	3,706	6,332	4,176	581	3,454	4,989
Region 8: Santa Ana	2,683	276	2,204	3,168	2,530	283	2,028	2,960
Region 9: San Diego	3,767	421	3,008	4,099	3,648	368	3,027	3,958
<i>Marginal enforcement costs (ψ), by regional water boards</i>								
Region 1: North Coast	1.18	0.49	0.50	1.98	0.38	0.15	0.14	0.64
Region 2: San Francisco Bay	0.92	0.19	0.38	1.00	0.45	0.25	0.10	1.00
Region 3: Central Coast	0.56	0.22	0.40	1.35	0.16	0.06	0.11	0.40
Region 4: Los Angeles	0.98	0.22	0.75	1.58	0.27	0.09	0.13	0.38
Region 5: Central Valley	0.77	0.31	0.35	1.78	0.17	0.06	0.08	0.31
Region 7: Colorado River	0.59	0.23	0.39	1.02	0.13	0.03	0.09	0.19
Region 8: Santa Ana	0.77	0.30	0.36	1.00	0.36	0.21	0.10	0.60
Region 9: San Diego	0.40	0.34	0.15	1.00	0.22	0.23	0.06	0.62

Notes: This table provides summary statistics of the estimated regulator preference parameters, before and after the 2006 institutional changes, for each regional water board.

the regulator preference estimates for the periods after the 2006 changes on the facility and local attributes. Tables A8 and A9 present the counterfactual analyses results, the counterparts of Tables 6 and 7 in Section 6.

E.5. Revisiting Disparities in Penalties and Regulator Preferences. Figure A3 shows the point-wise differences between the 5th and 95th percentiles of penalty schedules across the 221 facilities active in the first quarter of 2005 under the baseline scenario and the median regulator scenario (top lines), the 10th and 90th percentiles (middle lines) and the 25th and 75th percentiles (bottom lines), before and after the 2006 institutional changes, respectively.

There are three notable patterns in Figure A3. First, the differences between the 5th and 95th percentiles under the median regulator scenario are smaller than those under the baseline scenario, both before and after the 2006 changes. These findings are not specific to the choice of percentiles: the 10th-90th ranges and the interquartile ranges are all smaller under the median regulator than under the baseline scenario. This is because the facility-specific marginal benefits of compliance and the expected penalties are positively correlated, as shown in Panel (A) of Figure A4.

TABLE A7. Explaining Regulators' Preferences: After the 2006 Changes

Dependent variables:	$\log \gamma$	$\log \psi$
<i>Facility attributes</i>		
Major facility	0.143*** (0.019)	-0.078 (0.092)
First operated in 1982-7	-0.005 (0.009)	0.046 (0.076)
First operated after 1987	-0.018 (0.018)	-0.468* (0.261)
Advanced or tertiary	0.008 (0.011)	-0.005 (0.069)
Capacity utilization > 87%	-0.033*** (0.012)	0.243*** (0.090)
Service pop. < 10K	-0.007 (0.021)	-0.092 (0.078)
Run by a special district	-0.192*** (0.010)	0.161*** (0.060)
High threat to water	0.061*** (0.011)	-0.012 (0.063)
<i>Weather & pollution</i>		
Precipitation > 75 th pct.	-0.016 (0.013)	0.105 (0.085)
Precipitation < 25 th pct.	0.021 (0.029)	-0.013 (0.199)
Swimmable	-0.059*** (0.018)	-0.058 (0.069)
<i>County attributes</i>		
Irrigation use > 67%	-0.048*** (0.014)	-0.075 (0.102)
Average household income > \$57K	0.306*** (0.013)	-0.477*** (0.101)
Population density > 722/mi ²	-0.100*** (0.011)	0.265** (0.109)
Population density < 80/mi ²	0.246*** (0.017)	0.184* (0.109)
Proposition 84 approval > 50%	0.116*** (0.012)	0.020 (0.105)
<i>Regional water board FE†</i>		
Region 2: San Francisco Bay	0.135*** (0.035)	0.380** (0.162)
Region 3: Central Coast	0.034 (0.031)	-0.295** (0.122)
Region 4: Los Angeles	-0.242*** (0.036)	-0.330*** (0.164)
Region 5: Central Valley	0.356*** (0.031)	-0.593*** (0.096)
Region 7: Colorado River	0.518*** (0.039)	-1.036*** (0.202)
Region 8: Santa Ana	0.098*** (0.034)	0.068 (0.207)
Region 9: San Diego	0.147*** (0.039)	-0.232 (0.345)
Constant	7.648*** (0.051)	-1.157*** (0.199)
Adjusted R ²	0.942	0.585

Notes: This table reports the OLS regression results of the logarithm of the estimated marginal compliance cost and the logarithm of the estimated regulator preferences for each of the 221 facilities active in the first quarter of 2005 on all facility attributes used in the estimation. Robust standard errors under the assumption that the estimated parameters are measured without error are in parenthesis; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Second, the decreases in these point-wise ranges under the median regulator scenario are not very large, regardless of the negligence level, before the 2006 institutional changes. To be specific, the ratio of the 5th-95th range under the median regulator scenario to that under the baseline scenario varies from 0.72 to 0.84.

Third, such decreases based on the regulator preference estimates after the 2006 institutional changes are relatively large. For example, the ratio of the 5th-95th range under the median regulator scenario to that under the baseline scenario varies from

TABLE A8. The Effects of Regulatory Discretion: Further Results

	Baseline (1)	Median (2)	Uniform (3)	Linear (4)	$\psi = 0$ (5)	Green (6)
Prior to the 2006 institutional changes						
<i>Violation frequency</i>						
Mean	1.082	1.141	1.016	1.018	0.833	0.529
SD	0.785	1.000	0.636	0.745	0.522	0.465
<i>Penalty</i>						
Mean	2,380.8	2,096.3	2,411.8	2,757.6	3534.7	4,205.6
Dispersion	1,121.9	999.5	-	1,370.5	1425.4	216.9
After the 2006 institutional changes						
<i>Violation frequency</i>						
Mean	0.838	0.801	0.810	0.820	0.799	0.605
SD	0.603	0.600	0.558	0.594	0.548	0.456
Proportional, Mean	-	-0.044	-0.033	-0.022	-0.047	-0.279
Proportional, SD	-	-0.006	-0.074	-0.015	-0.092	-0.244
<i>Penalty</i>						
Mean	2,151.4	2,070.1	2,288.1	2,324.2	2,577.4	2,986.8
Dispersion	1,035.8	513.7	-	988.2	810.0	125.9
Proportional, Mean	-	-0.038	0.064	0.080	0.198	0.388
Proportional, Dispersion	-	-0.504	-1.000	-0.046	-0.218	-0.878

Notes: This table presents the summary statistics on compliance and enforcement under the baseline and the five counterfactual scenarios that reduce the regulator's discretion in Section 6.5. See Table 6 for the proportional changes compared to the baseline outcomes, based on the estimates for the period prior to the 2006 institutional changes.

0.38 to 0.45. This is rationalized by the fact that the correlation between the facility-level marginal benefits of compliance and the expected penalties is larger before the 2006 changes than after them, as can be seen in Panel (A) of Figure A4. In a similar vein, the correlation between the regulator marginal benefit estimates and the facility marginal cost estimates decreases after the 2006 changes (Panel (B)).

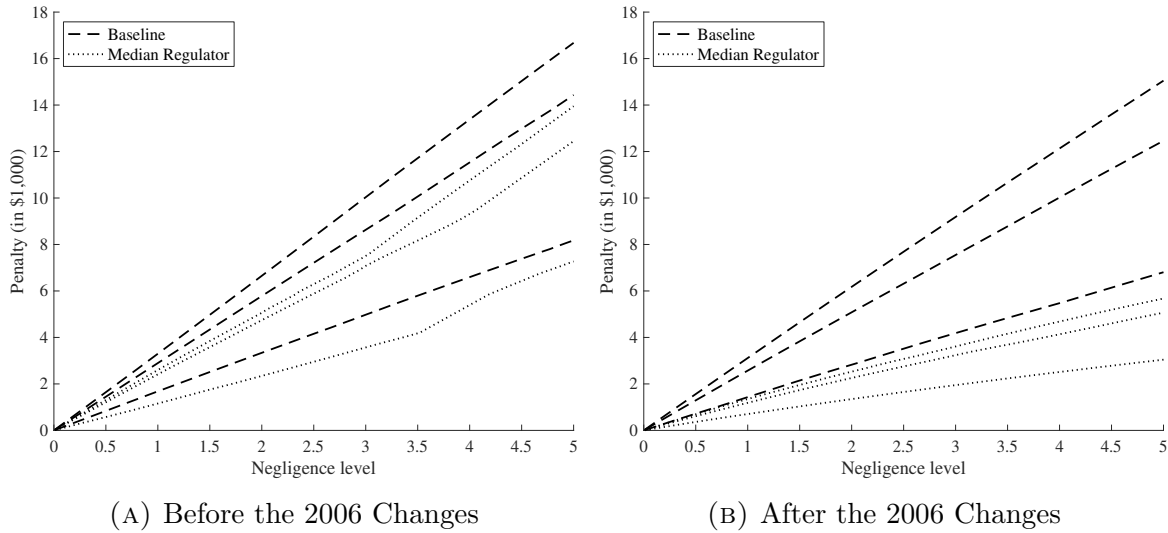
E.6. Regulator Costs. We compute the regulator's expected costs, as specified in (4), for each facility, conditional on the attributes, $\mathbf{x}_{i,t}$. The regulator's costs consist of three components. One such component is the compliance costs:

$$\int_0^{\bar{\theta}(\mathbf{x}_{i,t})} \theta \left\{ b[\bar{a}(\mathbf{x}_{i,t})|\mathbf{x}_{i,t}] - b[a^*(\theta|\mathbf{x}_{i,t})|\mathbf{x}_{i,t}] \right\} f(\theta|\mathbf{x}_{i,t}) d\theta. \quad (\text{A.9})$$

Another component is the external costs of violations:

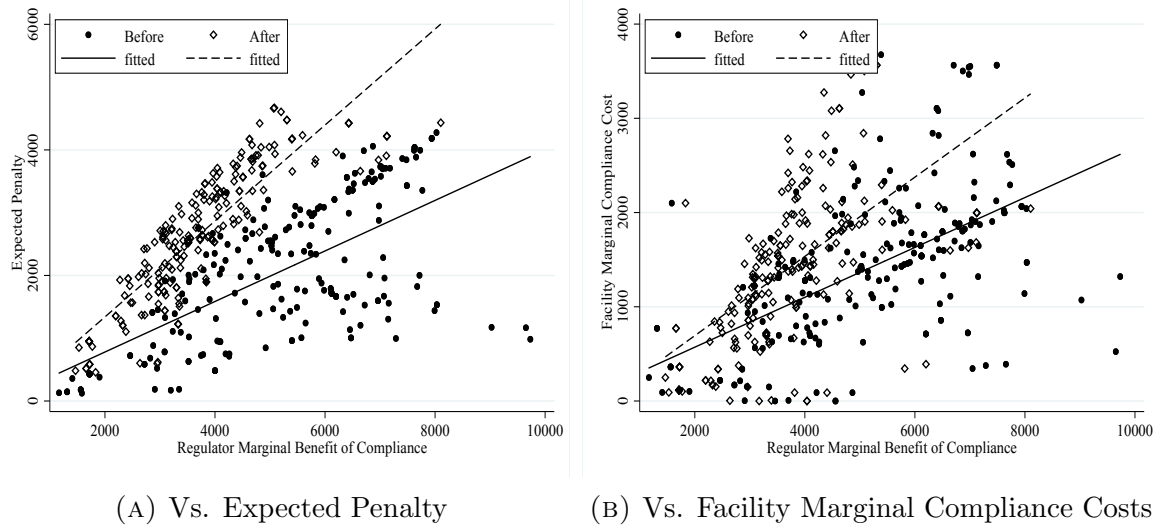
$$\gamma_j(\mathbf{x}_{i,t}) \int_0^{\bar{\theta}(\mathbf{x}_{i,t})} a^*(\theta|\mathbf{x}_{i,t}) f(\theta|\mathbf{x}_{i,t}) d\theta, \quad (\text{A.10})$$

FIGURE A3. Dispersion of Penalty Schedules



Notes: Panel (A) shows the point-wise differences between the 5th and 95th percentiles of penalty schedules across the 221 facilities active in the first quarter of 2005 under the baseline scenario and the median regulator scenario (top lines), the 10th and 90th percentiles (middle lines), and the 25th and 75th percentiles (bottom lines), before the 2006 institutional changes. Panel (B) is for the periods after the 2006 changes.

FIGURE A4. Disparities in Penalties and Regulator Preferences: Revisited



Notes: Panel (A) shows the scatter plot of the estimated marginal benefit of compliance as perceived by the regulator for each facility at the average negligence level before the 2006 institutional changes ($\bar{a} = 1.08$) and the expected penalty at the same negligence level. Panel (B) shows the scatter plot of the estimated marginal benefit and cost of compliance for each facility.

TABLE A9. Heterogenous Effects of Discretion: After the 2006 Changes

Dependent Variable:		<i>Any increase in the negligence level due to a change to the one-size-fits-all policy?</i>	
<i>Facility attributes</i>		<i>County attributes</i>	
Major facility	0.255*** (0.059)	Irrigation use > 67%	-0.290*** (0.069)
Started in 1982-87	-0.074 (0.052)	High income > \$57K	0.304*** (0.062)
Started after 1987	-0.0719 (0.088)	High density > 722/mi ²	-0.132** (0.057)
Advanced/tertiary	0.124*** (0.043)	Low density < 80/mi ²	0.387*** (0.078)
Cap. util. > 87%	0.017 (0.043)	High approval > 50%	0.374*** (0.071)
Service pop. < 10K	0.0002 (0.055)	<i>Regional water board FE†</i>	
Special district	-0.060 (0.039)	San Francisco Bay	0.437*** (0.126)
High threat to water	0.174*** (0.052)	Central Coast	0.062 (0.116)
<i>Weather & pollution</i>		Los Angeles	-0.581*** (0.134)
Precip. > 75 th pct.	-0.062 (0.056)	Central Valley	0.295*** (0.092)
Precip. < 25 th pct.	0.392* (0.199)	Colorado River	0.280 (0.196)
Swimmable	0.058 (0.073)	Santa Ana	-0.318** (0.133)
Constant	-0.222 (0.180)	San Diego	0.319* (0.193)
Adjusted R ²	0.727		

Notes: This table reports the OLS regression results where the dependent variable indicates if the facility would increase its negligence level (and hence increase the frequency of violations) under an alternative scenario compared to the baseline policy after the 2006 institutional changes. The unit of observation is each of the 221 facilities active in the first quarter of 2005. Robust standard errors under the assumption that the estimated parameters are measured without error are in parenthesis; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † Region 1 (North Coast) is omitted.

for $j \in \{pre, post\}$. The third component is the enforcement costs:

$$\begin{aligned}
& \psi_j(\mathbf{x}_{i,t}) \int_0^{\bar{\theta}(\mathbf{x}_{i,t})} e^* [a^*(\theta|\mathbf{x}_{i,t})|\mathbf{x}_{i,t}] f(\theta|\mathbf{x}_{i,t}) d\theta \\
&= \psi_j(\mathbf{x}_{i,t}) \int_0^{\bar{\theta}(\mathbf{x}_{i,t})} \left(\theta b[a^*(\theta|\mathbf{x}_{i,t})|\mathbf{x}_{i,t}] - \int_0^{\bar{\theta}(\mathbf{x}_{i,t})} b[a^*(v|\mathbf{x}_{i,t})|\mathbf{x}_{i,t}] dv \right) f(\theta|\mathbf{x}_{i,t}) d\theta \\
&= \psi_j(\mathbf{x}_{i,t}) \int_0^{\bar{\theta}(\mathbf{x}_{i,t})} \left(\int_0^{a^*(\theta|\mathbf{x}_{i,t})} \theta^*(a|\mathbf{x}_{i,t}) b'(a|\mathbf{x}_{i,t}) da \right) f(\theta|\mathbf{x}_{i,t}) d\theta, \tag{A.11}
\end{aligned}$$

for $j \in \{pre, post\}$, where $\theta^*(\cdot|\mathbf{x}_{i,t})$ is the inverse of $a^*(\cdot|\mathbf{x}_{i,t})$. The first equality in (A.11) follows from (7), and the second equality from integration by parts and change of variables. We estimate (A.9)–(A.11) for each facility, using our estimates of $b'(\cdot|\mathbf{x}_{i,t})$, $a^*(\cdot|\mathbf{x}_{i,t})$, and $f(\theta|\mathbf{x}_{i,t})$. We approximate $\bar{\theta}(\mathbf{x}_{i,t})$ and $\bar{a}(\mathbf{x}_{i,t})$ by the ninety-ninth percentile of the estimated distributions of types and negligence levels, conditional on $\mathbf{x}_{i,t}$, respectively.

Table A10 presents the average values of the estimates of (A.9)–(A.11) across the 221 facilities in our sample. For a fair comparison of the regulator costs across the

TABLE A10. Regulator Expected Costs

	Compliance	External	Enforcement	Total
(1) Baseline	2,539.2	4,734.8	1,816.6	9,090.7
(2) Median regulator	2,378.3	5,369.5	1,647.6	9,395.5
(3) Uniform penalty	2,783.4	4,505.6	1,954.6	9,243.5
(4) Linear penalty	2,703.5	4,434.8	2,143.6	9,282.0
(5) No enforcement costs	3,429.7	3,534.6	2,820.5	9,834.8
(6) Green regulator	4,902.6	2,474.5	3,306.7	10,683.8

Notes: This table presents the average value across facilities of each component of the regulator's expected costs for the baseline and the five counterfactual scenarios of Table 6. The table reports the results based on the estimates for the period prior to the 2006 institutional changes.

baseline and the counterfactual scenarios, we use the estimated regulator preference parameters $\gamma_{pre}(\mathbf{x}_{i,t})$ and $\psi_{pre}(\mathbf{x}_{i,t})$ in all scenarios. For this reason, for example, the enforcement costs in Row (5) are nonzero.

We find that the policies reducing regulatory discretion in comparison to the baseline scenario—namely, the uniform policy and the linear policy (Rows (3) and (4) of Table A10)—would increase the regulator's expected costs by about 2-3 percent. In the green regulator scenario, facilities would increase compliance by 51 percent (Row (5) of Table 6) at the expense of doubling compliance and enforcement costs (Row (6) of Table A10), relative to the baseline scenario. However, external costs would decrease by half, and the total expected regulator costs would increase by 18 percent.

E.7. Extra Counterfactual Analyses. Table A11 shows the results of five additional counterfactual analyses that are not included in Table 6. Row (1) presents the equilibrium violation frequency and penalty statistics in the scenario in which facilities face a hypothetical regulator whose assessment of external violation costs is the median value of γ for all facilities, but who continues to recognize the different enforcement costs at the estimated ψ values for each facility. The regulator considered in Row (2) is the opposite of that of Row (1): her enforcement cost is the median value of ψ for all facilities, but she perceives external violation costs at the estimated value of γ for each facility. We find that the dispersion of the equilibrium penalty stringency, as measured by the standard deviation of the expected penalties across the facilities for the average negligence level ($\bar{a} = 1.08$), decreases under either regime, relative to the baseline case. But the decrease in the Row (2) scenario, 8.8 percent, is almost twice as large as that in Row (1), 4.5 percent. These results suggest

TABLE A11. Extra Counterfactual Analyses

	Violation Frequency		Penalty	
	Mean	SD	Mean	Disparity
(1) Homogenous γ	0.113	0.322	-0.081	-0.045
(2) Homogenous ψ	0.068	-0.005	-0.015	-0.088
(3) Sub-optimal tailored	-0.061	0.060	-0.012	-0.153
(4) Sub-optimal linear	0.000	0.000	0.101	-0.106
(5) Doubled enforcement costs	0.238	0.239	-0.178	-0.001

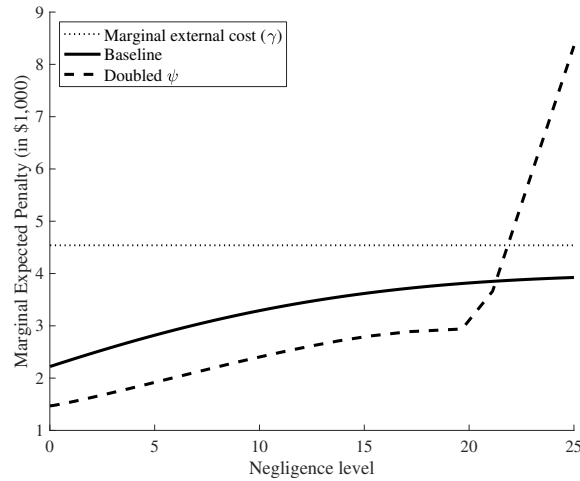
Notes: This table presents the results of five extra counterfactual scenarios. For each scenario, we report the proportional changes, relative to the baseline scenario, in the following four statistics: the mean and the standard deviation of the equilibrium violation frequency across facilities; the mean of the equilibrium expected penalties; and the standard deviation of the equilibrium penalties at an identical negligence level ($a = 1.08$) across facilities, as a measure of disparities in equilibrium penalty stringency. In Row (1), every facility is under a regulator whose marginal external cost of violations (γ) is at the median across all facilities, while her marginal enforcement costs (ψ) vary across facilities at the estimated values. The regulator considered in Row (2) is the opposite: the regulator's ψ is identical across all facilities, at the median value, and her γ varies, at the estimated values. The penalty schedule of Row (3) achieves the same average violation frequency as the uniform policy, but is tailored to facility and local attributes. In Row (4), the penalty schedule is a linear penalty schedule designed to induce each facility to choose the same negligence level as the baseline. In the scenario in Row (5), the marginal enforcement costs ψ are doubled. The table reports the results based on the estimates for the period prior to the 2006 institutional changes.

that heterogeneity in enforcement costs are more important than variation in external violation costs in explaining penalty disparities.

To illustrate the benefits for the regulator of being able to individually consider the attributes of each facility, we consider penalty schedules that (i) are tailored to facility and local attributes, and (ii) would achieve the same average violation frequency across the facilities as the uniform policy considered in Row (2) of Table 6, in Section 6.5.1. There may be many such schedules, and we consider a particularly simple example, as described in Appendix D.1. Row (3) of Table A11 presents the results. The mean penalties would decrease by 1.2 percent compared to the baseline scenario, and would be lower than the mean penalties under the uniform policy by $|(1 - 0.012)/1.013 - 1| \approx 2.5$ percent.

Row (4) of Table A11 provides the results of a counterfactual scenario in which each facility faces the linear penalty that would lead to the same expected number of violations as in the baseline scenario. The details on the implementation of this counterfactual are in Appendix D.2. We find that the mean penalties would increase

FIGURE A5. Enforcement Costs and Marginal Penalties



Notes: This graph presents the equilibrium marginal expected penalties, $e'(\cdot)$, as a function of the level of negligence, for a facility whose marginal external cost is at the 75th percentile (\$4,550, indicated as a dotted line). The solid line is for the the baseline prior to the 2006 changes, and the dashed line is for the case where the marginal enforcement cost (ψ) for that facility is doubled.

by 10 percent compared to the baseline scenario, although the negligence level induced by either policy is the same for all facilities.

As discussed in 4.3, the presence of enforcement costs, coupled with the information asymmetry between a regulator and a facility, creates a gap between the first-best and the equilibrium negligence levels. To assess the importance of enforcement costs, we also consider a regulator that has higher enforcement costs than in the baseline scenario. Specifically, for each facility, the regulator considered in Row (5) of Table A11 has the same marginal external costs of violations (γ) as in the baseline case, but her enforcement costs (ψ) are doubled. Higher enforcement costs would increase the equilibrium level of violation frequency, by 24 percent compared to the baseline scenario, and reduce the equilibrium penalties by 18 percent. Figure A5 shows that this increase in the equilibrium violation frequencies is driven by a lower marginal penalty for relatively mild offense, despite a steeply higher marginal penalty for serious offenses, relative to the baseline scenario. The figure presents, for a facility, the equilibrium marginal expected penalties as a function of the level of negligence, $e'(\cdot)$, under both the baseline and the doubled- ψ scenarios. This pattern is not limited to the particular facility shown in the figure, and illustrates the regulator's optimal response in designing the penalty schedule to an increase in enforcement costs, as emphasized by Mookherjee and Png (1994).

APPENDIX F. SENSITIVITY ANALYSES

This Appendix assesses the sensitivity of our findings to some of the assumptions made in our main empirical analysis. First, we develop and estimate an extension of the model that explicitly incorporates information about violation severity. Here, the negligence level set by a facility affects not only the number but also the severity of violations, and the regulator perceives the social costs of severe and non-severe violations differently. In further sensitivity analyses, we consider alternative choices on the length of the penalty window, the sample selection, and the covariates.

F.1. Alternative Specification 1: Endogenous Violation Severity.

F.1.1. *Model.* As in the model presented in the main text, a facility has privately known type θ , drawn from a distribution $F(\cdot)$. The facility sets a negligence level a , which affects the frequency and the severity of violations. Differently from the model in the main text, each occurred violation may be *severe* or not. Denote by $k \in \mathbb{N}$ and $m \in \mathbb{N}$, respectively, the total number of violations and the number of severe violations by a facility in a period. Also, let $\iota \in \mathbb{R}$ be a random variable representing features of the occurred violations that are not captured by k and m . We make the following assumption concerning the relationship between negligence levels and violation outcomes:

ASSUMPTION A1. $k \sim \text{Poisson}(a)$; $m|k \sim \text{Binomial}(k, s(a))$; and $\iota \perp a$.

After the violations occur, (k, m, ι) are observed by both the regulator and the facility. Hence the penalty schedule is a function of (k, m, ι) , which we denote by $\epsilon(\cdot, \cdot, \cdot)$. Let $e(a) \equiv \mathbb{E}[\epsilon(k, m, \iota)|a]$. Like in the main text, we assume that the regulator chooses $e(\cdot)$, since, for any continuous $e(\cdot)$, there exists an enforcement schedule $\epsilon(\cdot, \cdot, \cdot)$ that implements $e(\cdot)$. We revisit this assumption below, in our discussion of identification. The facility derives compliance costs $\theta[b(\bar{a}) - b(a)]$ from the negligence level a . The facility's objective function is thus given by (2). We allow the social costs of violations, as perceived by the regulator, to depend on (k, m, ι) . Specifically, let us denote these costs by $c(k, m, \iota)$. The total regulator's expected costs are

$$\int_0^{\bar{\theta}} \left(\theta \{b(\bar{a}) - b[a(\theta)]\} + \mathbb{E}[c(k, m, \iota)|a(\theta)] + \psi e[a(\theta)] \right) f(\theta) d\theta.$$

By defining $h(a) \equiv \mathbb{E}[c(k, m, \iota)|a]$, we can write the regulator's objective function as (4). Under Assumptions 1 and 2, the optimality conditions derived in the main text still hold in the extended model.

F.1.2. *Identification.* In addition to the observables described in the main text, we assume that the data report the severity of each occurred violation. In our estimation, we use an indicator of whether the violation is classified by the water boards as a *priority*, as explained in the notes in Table 2, to measure severity. Like in the main text, we consider a vector $\mathbf{x}$ of observed facility attributes, and thus all primitives and the equilibrium objects of the model are allowed to vary with $\mathbf{x}$. The regulator may implement the expected enforcement function $e(\cdot; \mathbf{x})$ using a variety of schedules $\epsilon(\cdot, \cdot, \cdot; \mathbf{x})$. In the following lemma, we consider a particular schedule:

LEMMA A2. *Let $e(a; \mathbf{x})$ be a continuous function with domain $a \in [0, \bar{a}]$, where $\bar{a} < \infty$, and fix any vector $\mathbf{x}$. Then, for any $\delta > 0$, there exists $\epsilon_1(\cdot, \cdot, \cdot; \mathbf{x}) : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ and $\epsilon_2(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ such that*

$$\sup_{a \in [0, \bar{a}]} |e(a; \mathbf{x}) - \mathbb{E}[\epsilon_1(k, m; \mathbf{x}) + \epsilon_2(\iota) | a, \mathbf{x}]| < \delta,$$

and $\mathbb{E}[\epsilon_2(\iota) | k, m, \mathbf{x}] = 0$ for any (k, m) .

Proof. Following the same argument as in Lemma A1, we can show that, for every continuous $e(\cdot; \mathbf{x})$, there is a function $\epsilon_1(\cdot, \cdot, \cdot; \mathbf{x})$ such that $e(a) \approx \mathbb{E}[\epsilon_1(k, m; \mathbf{x}) | a, \mathbf{x}]$ for any $a \in [0, \bar{a}]$. Given that, an alternative schedule, $\epsilon(k, m, \iota, \mathbf{x}) \equiv \epsilon_1(k, m, \mathbf{x}) + \epsilon_2(\iota)$ such that $\mathbb{E}[\epsilon_2(\iota) | a, \mathbf{x}] = 0$ implements $e(\cdot; \mathbf{x})$ as well. Under the assumption that $\iota \perp a$ (Assumption A1), $\mathbb{E}[\epsilon_2(\iota) | a, \mathbf{x}] = \mathbb{E}[\epsilon_2(\iota) | \mathbf{x}]$. Therefore, any $\epsilon_2(\cdot)$ that satisfies $\mathbb{E}[\epsilon_2(\iota) | k, m, \mathbf{x}] = 0$, along with $\epsilon_1(\cdot, \cdot, \cdot; \mathbf{x})$, also implements $e(\cdot; \mathbf{x})$. $\square$

Given the lemma, we make the following assumption:

ASSUMPTION A2. *The regulator, among the schedules that implement $e(a; \mathbf{x})$ for any $a \in [0, \bar{a}]$ and $\mathbf{x}$, employs a schedule of the following form, $\epsilon(k, m, \iota; \mathbf{x}) \equiv \epsilon_1(k, m; \mathbf{x}) + \epsilon_2(\iota)$, with $\mathbb{E}[\epsilon_2(\iota) | k, m, \mathbf{x}] = 0$ for any (k, m) .*

Under this assumption, as k , m , $\mathbf{x}$, and the assigned penalties are observable, $\epsilon_1(k, m, \mathbf{x})$ is directly identified from the data—and so is $e(\cdot; \mathbf{x})$. Regarding the distribution of negligence, Lemma 1 still holds under Assumption A1, which allows us to apply the same strategy as in the main text for the identification of $G_j(\cdot; \mathbf{x})$.

The additional primitive to be recovered in the extended model is $s(\cdot)$, the probability that each occurred violation is severe, as a function of the negligence level a . We impose the following parametric assumption concerning this function:

ASSUMPTION A3. *The probability $s(a)$ that an occurred violation is severe, conditional on the facility setting a negligence equal to a , is $\Phi(p_0 + p_1 a)$, where $\Phi(\cdot)$ is the standard normal CDF.*

Under Assumption A3, the identification of $s(\cdot)$ consists of identifying the parameters p_0 and p_1 , which we achieve by exploiting the relationship between the proportion of severe violations and the negligence distribution in the pre- and post-2006 periods. Recall that the penalty schedules $\epsilon_{1,j}(k, m, \mathbf{x})$ for $j \in \{pre, post\}$ are directly identified from the data. Having identified p_0, p_1 , we identify $e_j(\cdot; \mathbf{x})$, the expected penalty as a function of negligence, as

$$e_j(a; \mathbf{x}) \equiv \sum_{k=0}^{\infty} \sum_{m=0}^k \frac{\exp(-a) \Phi[p_0 + p_1 a]^m \{1 - \Phi[p_0 + p_1 a]\}^{k-m}}{\zeta!(k - \zeta!)} \epsilon_{1,j}(k, m, \mathbf{x}).$$

The rest of the identification argument follows the steps from the main text. Under Assumptions 1-3, Proposition 2 holds. Under Assumptions 1-5, Proposition 3 is valid.

F.1.3. *Estimation.* We adopt the same parametric assumptions on the distribution of negligence as in the basic model: The number of violations by a facility with covariates $\mathbf{x}$ in period $j \in \{pre, post\}$ follows a negative binomial distribution with mean $\exp(\beta_{0,j} + \beta_1 \mathbf{x})$ and variance $\exp(\beta_{0,j} + \beta_1 \mathbf{x}) [1 + \Delta(\mathbf{z})^{-1} \exp(\beta_{0,j} + \beta_1 \mathbf{x})]$, where the term $\Delta(\mathbf{z}) = \exp(\mathbf{z}\delta)$ and $\mathbf{z}$ is a subset of $\mathbf{x}$. Define $\beta_j \equiv (\beta_{0,j}, \beta_1)$, for $j \in \{pre, post\}$, and let $g(\cdot | \mathbf{x}; \delta, \beta_j)$ be the density function of negligence levels implied by the parameters δ, β_{pre} and β_{post} , conditional on $\mathbf{x}$.

We estimate the severity parameters p_0 and p_1 by maximum likelihood. Specifically, under Assumptions A1 and A3, we can write the joint distribution of (k, m) , conditional on the negligence level a , as

$$\Pr(k, m | a; p_0, p_1) = \frac{a^k \exp(-a)}{k!} \binom{k}{m} \Phi(p_0 + p_1 a)^m [1 - \Phi(p_0 + p_1 a)]^{k-m}.$$

We can then write the joint distribution of (k, m) for a facility, conditional on $\mathbf{x}$, as

$$\Pr(k, m; p_0, p_1, \delta, \beta_j, \mathbf{x}) = \int \Pr(k, m | a; p_0, p_1) g(a | \mathbf{x}; \delta, \beta_j) da,$$

for $j \in \{pre, post\}$. Denote by $\hat{\delta}$, $\hat{\beta}_{pre}$ and $\hat{\beta}_{post}$, respectively, the estimates of δ , β_{pre} and β_{post} presented in the main text. Under the assumptions above, these estimates are consistent. A maximum likelihood estimator of p_0 and p_1 is then

$$(\hat{p}_0, \hat{p}_1) = \arg \max_{p_0, p_1} \sum_t \sum_i \log \left[\int \Pr(k_{i,t}, m_{i,t} | a; p_0, p_1) g(a | \mathbf{x}_{i,t}; \hat{\delta}, \hat{\beta}_j) da \right],$$

where $m_{i,t}$ is the number of severe violations by facility i in period t . To estimate the enforcement functions, we extend specification (13) as follows:

$$\begin{aligned}\eta_{1i,t}^* &= \mathbf{x}_{i,t}\phi_{1,x} + \mathbb{1}_{\{t>2006\}}\phi_{1,post} + \phi_{1,k}k_{i,t} + \phi_{1,m}m_{i,t} + u_{1i,t}, \\ \log \eta_{2i,t}^* &= \log \left[\exp(\mathbf{x}_{i,t}\phi_{2,x}) k_{i,t} + \phi_{2,k^2}k_{i,t}^2 + \phi_{2,m}m_{i,t} \right] + u_{2i,t}, \\ \text{and } \epsilon_{i,t} &= \begin{cases} \eta_{2i,t}^* & \text{if } \eta_{1i,t}^* \geq 0, \\ 0 & \text{otherwise,} \end{cases}\end{aligned}\tag{A.12}$$

where $(u_{1i,t}, u_{2i,t})$'s are i.i.d. draws from a bivariate normal distribution with mean zero, independent of $(k_{i,t}, m_{i,t}, \mathbf{x}_{i,t})$. By Assumption A2, we have that $\epsilon_{1,pre}(k, m, \mathbf{x}) = \mathbb{E}[\epsilon_{i,t}|k, m, \mathbf{x}, t < 2006]$ and $\epsilon_{1,post}(k, m, \mathbf{x}) = \mathbb{E}[\epsilon_{i,t}|k, m, \mathbf{x}, t > 2006]$. Therefore, the difference between the observed penalty amount and the predicted penalty amount—i.e., $(\epsilon_{i,t} - \epsilon_{1,j}(k, m, \mathbf{x}))$, where $j = pre$ for $t < 2006$ and $j = post$ for $t > 2006$ —is $\epsilon_2(\epsilon_{i,t})$, or the part of the enforcement schedule associated with the violation aspects that are not observed in the data. With the parameters in (A.12), denoted by ϕ , we compute $\epsilon_{1,j}(k, m, \mathbf{x}; \phi)$ for $j \in \{pre, post\}$. We obtain estimates $\hat{\phi}$ by maximum likelihood. We then estimate the expected penalty for a facility with covariates $\mathbf{x}$, as a function of the negligence level a , by

$$\hat{e}_j(a|\mathbf{x}) \equiv \exp(-a) \sum_{k=0}^{\infty} \sum_{m=0}^k \frac{\Phi(\hat{p}_0 + \hat{p}_1 a)^m [1 - \Phi(\hat{p}_0 + \hat{p}_1 a)]^{k-m}}{\zeta!(k - \zeta!)} \epsilon_{1,j}(k, m, \mathbf{x}; \hat{\phi}).$$

Having estimates for the enforcement function and the distribution of negligence, we proceed with the estimation of the model primitives, as in the main text.

F.2. Other Sensitivity Analyses. We consider four other alternative specifications of our model. In *alternative specification 2*, we estimate the penalty schedule by considering all penalties within three years of the occurrence of each violation, as opposed to the four years used in the original results. This change allows us to use a longer period of data for the penalty schedule estimation in the first step. Specifically, we employ the penalties for the violations of 2000–2002 (2009–2011) to estimate the penalty schedule before (after) the 2006 institutional changes, instead of 2000–2001 (2009–2010), as in the original results.

In *alternative specification 3*, we change the unit of observation from a facility-quarter to a facility-semester. For a given period, a facility draws its cost type and determines the level of negligence; and a regulator sets a penalty schedule over the violations during the period. A quarter reflects the distinct precipitation patterns

across four seasons in California. A semester, however, may also be a suitable period to consider, because, among all 1,605 penalty actions imposed on domestic wastewater treatment facilities in 2000-2014, the median period of violations comprised by an unique penalty action (either an ACL or a settlement in court) is 7 months.

The results presented in the main text are based on estimates of the penalty schedule in which we truncate facility-quarter years with more than 25 violation. *Alternative specification 4* and *alternative specification 5* address the estimation of the model with truncations at 20 and 30 violations, respectively.

Finally, *alternative specification 6* considers a more flexible estimator of the negligence distribution. This specification incorporates into $\mathbf{z}$, the vector of covariates that determines the overdispersion coefficient $\Delta(\mathbf{z})$, all the elements of $\mathbf{x}$ used in the estimation of other parts of the model.

The results from alternative specification 1-6 are in Tables A12 and A13. All the main results are similar to our original estimates. Specifically, the marginal compliance cost estimates and regulator preferences are very similar to those presented in the main text. Perhaps most importantly, the main results of our counterfactual analysis in Section 6.5 are robust to these six alternative specifications.

TABLE A12. Sensitivity Analyses: Model Primitive Estimates

	Sensitivity: Primitives				
	Marginal Compliance Cost	Environmental Cost per Violation (γ)		Enforcement Cost per Penalty (ψ)	
		Before	After	Before	After
<i>Original specification</i>					
Median	1431.0	3589.4	3157.1	0.865	0.204
Interquartile range	1090.3	1633.0	916.5	0.483	0.220
<i>Alternative specification 1: Endogenous severity</i>					
Median	1401.4	3517.1	3054.9	0.712	0.184
Interquartile range	1086.1	1525.5	959.6	0.545	0.215
<i>Alternative specification 2: Penalty within 3 years</i>					
Median	1324.5	3315.1	3042.2	0.922	0.237
Interquartile range	1160.5	1192.8	861.9	0.557	0.256
<i>Alternative specification 3: Semester-long period</i>					
Median	1252.8	3717.1	3215.7	0.636	0.130
Interquartile range	1261.2	1277.9	972.5	0.652	0.130
<i>Alternative specification 4: Violations truncated at 20</i>					
Median	1438.7	3856.5	3308.3	0.999	0.235
Interquartile range	1015.9	1927.6	850.9	0.367	0.229
<i>Alternative specification 5: Violations truncated at 30</i>					
Median	1454.5	3455.6	3114.6	0.696	0.174
Interquartile range	1094.3	1639.4	957.6	0.585	0.219
<i>Alternative specification 6: Flexible overdispersion</i>					
Median	1024.9	3608.5	3194.2	0.702	0.185
Interquartile range	1379.1	1480.4	954.3	0.551	0.214

TABLE A13. Sensitivity Analyses: Counterfactuals

	Median Regulator (1)	Uniform Penalty (2)	Linear Penalty (3)	$\psi = 0$ (4)	Green Regulator (5)
Equilibrium violation frequency					
<i>Original specification</i>					
Proportional change in Mean	0.055	-0.062	-0.064	-0.230	-0.511
Proportional change in SD	0.274	-0.191	-0.048	-0.335	-0.408
<i>Alternative specification 1: Endogenous severity</i>					
Proportional change in Mean	-0.015	-0.071	-0.053	-0.284	-0.553
Proportional change in SD	0.192	-0.179	-0.037	-0.356	-0.440
<i>Alternative specification 2: Penalty within 3 years</i>					
Proportional change in Mean	0.047	-0.062	-0.053	-0.209	-0.491
Proportional change in SD	0.161	-0.240	-0.065	-0.289	-0.463
<i>Alternative specification 3: Six months-long period</i>					
Proportional change in Mean	-0.010	-0.022	-0.035	-0.185	-0.558
Proportional change in SD	0.088	-0.133	-0.040	-0.211	-0.528
<i>Alternative specification 4: Violations truncated at 20</i>					
Proportional change in Mean	0.077	-0.062	-0.071	-0.246	-0.510
Proportional change in SD	0.275	-0.203	-0.067	-0.337	-0.426
<i>Alternative specification 5: Violations truncated at 30</i>					
Proportional change in Mean	-0.002	-0.075	-0.051	-0.213	-0.508
Proportional change in SD	0.200	-0.194	-0.040	-0.331	-0.398
<i>Alternative specification 6: Flexible overdispersion</i>					
Proportional change in Mean	0.051	-0.168	-0.043	-0.173	-0.483
Proportional change in SD	0.173	-0.267	-0.013	-0.269	-0.406
Equilibrium penalty					
<i>Original specification</i>					
Proportional change in Mean	-0.063	-0.072	0.527	0.485	3.081
Proportional change in Disparity	-0.110	-1.000	-0.070	0.271	-0.807
<i>Alternative specification 1: Endogenous severity</i>					
Proportional change in Mean	0.024	-0.048	0.450	0.465	2.460
Proportional change in Disparity	-0.170	-1.000	0.245	0.155	-0.917
<i>Alternative specification 2: Penalty within 3 years</i>					
Proportional change in Mean	-0.134	-0.037	0.507	0.620	2.624
Proportional change in Disparity	-0.228	-1.000	-0.059	-0.098	-0.774
<i>Alternative specification 3: Six months-long period</i>					
Proportional change in Mean	-0.003	0.046	0.543	0.632	4.489
Proportional change in Disparity	-0.183	-1.000	-0.031	0.140	-0.604
<i>Alternative specification 4: Violations truncated at 20</i>					
Proportional change in Mean	-0.063	-0.051	0.569	0.715	3.203
Proportional change in Disparity	-0.021	-1.000	0.230	0.288	-0.747
<i>Alternative specification 5: Violations truncated at 30</i>					
Proportional change in Mean	-0.016	-0.070	0.393	0.708	2.964
Proportional change in Disparity	-0.179	-1.000	0.149	0.279	-0.846
<i>Alternative specification 6: Flexible overdispersion</i>					
Proportional change in Mean	-0.024	-0.039	0.433	0.735	2.880
Proportional change in Disparity	-0.133	-1.000	0.166	0.184	-0.774

REFERENCES

- Blundell, Wesley, Gautam Gowrisankaran, and Ashley Langer. 2020. "Escalation of Scrutiny: The Gains from Dynamic Enforcement of Environmental Regulations." *American Economic Review* 110 (8):2558–2585.
- California State Water Resources Control Board. 2010. "Deconstructing Enforcement: A Primer on Water Quality Enforcement." Tech. rep., Office of Enforcement,. Available at https://www.waterboards.ca.gov/water_issues/programs/enforcement/docs/deconstructing_enforce2010aug.pdf (last accessed in August 2019).
- Cameron, A Colin and Pravin K Trivedi. 2013. *Regression Analysis of Count Data*. Cambridge University Press.
- Duflo, Esther, Michael Greenstone, Rohini Pande, and Nicholas Ryan. 2018. "The Value of Regulatory Discretion: Estimates from Environmental Inspections in India." *Econometrica* 86 (6):2123–2160.
- Harrington, Winston. 1988. "Enforcement Leverage When Penalties are Restricted." *Journal of Public Economics* 37 (1):29–53.
- Mookherjee, Dilip and I. P. L. Png. 1994. "Marginal Deterrence in Enforcement of Law." *Journal of Political Economy* 102 (5):1039–1066.
- Royden, Halsey Lawrence and Patrick Fitzpatrick. 1988. *Real Analysis*. Macmillan New York.
- US Environmental Protection Agency. 2004. "Primer for Municipal Wastewater Treatment Systems." Tech. rep., Office of Wastewater Management, Office of Water, EPA 832-R-04-001,. Available at <https://www3.epa.gov/npdes/pubs/primer.pdf> (last accessed in August 2019).
- Weirich, Scott R, JoAnn Silverstein, and Balaji Rajagopalan. 2015. "Resilience of Secondary Wastewater Treatment Plants: Prior Performance is Predictive of Future Process Failure and Recovery Time." *Environmental Engineering Science* 32 (3):222–231.